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Chapter 3: The Macro-Cognitive Accounting Framework (MCAF)

Abstract

The previous chapter argued that the clock is not time. But civilization cannot avoid using clocks. Tracking the gap between the clock and existence requires a bookkeeping structure. This chapter establishes that structure.

Although computing power has significantly extended atmospheric predictability, it remains constrained by the limits of chaos; meanwhile, algorithmic interventions in human cognition and behavior have already produced observable effects. The root of this asymmetry lies not in the cognitive system being simpler, but in the fact that its defense mechanisms rely on a structural friction that requires continuous work to sustain—once the work ceases, the difference between the projection side and the defense side precipitates as cognitive debt.

This chapter establishes the Macro-Cognitive Accounting Framework (MCAF), using a six-variable meta-formula to model the projection-defense dynamic as an interdisciplinary chart of accounts—its mathematical identity is a classificatory discipline, not a solvable dynamical system; the mathematical concretization of the variables is left to empirical mapping in application scenarios (§7.2). Both the projection and defense sides use product-form aggregation. Their asymmetry does not arise from multiplication itself, but from the variables' different growth and constraint conditions: holding the other variables fixed, unilateral growth in any projection-side variable raises the total load, whereas effective defense requires all three defense-side variables to remain operative. The three clocks mark temporal asymmetry through characteristic periods: the techno-capital period is generally shorter than the governance period, while the delay from debt sedimentation

to first manifestation and the governance response are not assigned a fixed ordering in advance. If debt manifests before the governance response is complete, the system enters what this chapter calls a crisis state.

There are three core judgments. Cognitive friction is not a design flaw to be eliminated—it is a necessary condition for maintaining agency. The decay of meta-audit capacity has two independent paths: the automation paradox renders operational friction obsolete, and algorithmic *xiangyuan* displaces affective friction. The two paths act in synergy: meta-audit capacity is not defeated but abandoned. The same structural act running throughout the entire chapter is discretization—discretization itself is not the lesion, but usurping the outcomes of discretized instrumental rationality as the substance of value rationality is the root cause of the erosion of the defense line.

Keywords

Macro-Cognitive Accounting Framework (MCAF); cognitive debt; meta-formula; meta-audit capacity; cognitive friction; cognitive Reynolds number; narrative leverage ratio; discretization

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§0 The Genesis of the Problem: Two Asymmetries

§0.1 What Computing Power Conquered, and What It Did Not

In April 1950, von Neumann, Charney, and others used the Electronic Numerical Integrator and Computer (ENIAC) at the Aberdeen Proving Ground to complete the first numerical weather prediction in human history—using approximately 24 hours of computing time to forecast 24 hours of weather (Charney, Fjørtoft & von Neumann, 1950). Von Neumann's ambition extended far beyond this. In 1955 he wrote: "But our knowledge of the dynamics and the controlling processes in the atmosphere is rapidly approaching a level that would permit such prediction. Probably intervention in atmospheric and climatic matters will come in a few decades, and will unfold on a scale difficult to imagine at present."

Over seventy years later, this engineering ambition has encountered fundamental limits in the face of the complexity of the physical world (as implied by the chaos limits of weather forecasting). Machine-learning research, by optimizing initial conditions, has extended deterministic forecast skill from roughly two weeks to beyond 30 days in hindcast experiments (Vonich & Hakim, 2026)—the two-week limit is thus no longer absolute, yet the deep difficulty revealed by Lorenz (1963) remains: minute differences in initial conditions can grow rapidly and constrain long-range prediction, and whether such initial conditions can be identified in real time for operational forecasting remains an open question. Computing power did not conquer the weather.

Yet the descendants of the same computational architecture have achieved progress far beyond expectations in another domain—algorithmic intervention in human cognition and behavior. Two independent experiments have confirmed the structural possibility of such intervention at the individual level and the system level, respectively. At the individual level, a pre-registered controlled experiment showed that when GPT-4 obtained basic demographic information about an opponent, its persuasiveness in multi-round debates significantly exceeded that of human opponents—in debate pairs where artificial intelligence (AI) and humans were not equally persuasive, personalized GPT-4 was more persuasive 64.4% of the time, with an 81.2% relative increase in the odds of higher post-debate agreement compared with the human–human baseline condition (Salvi et al., 2025). At the

system level, a randomized experiment independent of the platform side found that X's (formerly Twitter) algorithmic recommendation feed significantly shifted users' policy attitudes in a conservative direction—random assignment indicates that the effect cannot be explained solely by users' self-selection into feed settings, and algorithm-induced changes in following behavior are an important mechanism of its persistence (Gauthier et al., 2026). It is worth noting the asymmetry of this effect: switching from a chronological to an algorithmic feed shifted political opinion towards more conservative positions, whereas switching in the opposite direction—from the algorithmic back to the chronological feed—had no comparable effects; the mechanism is that exposure to algorithmic content leads users to follow conservative political activist accounts, which they continue to follow even after switching off the algorithm, so the effect of initial exposure persists after algorithmic curation is removed. This on-with-effect, off-with-no-reversal asymmetry shows that algorithm-induced following behavior can produce persistent effects. It therefore supplies a candidate for scenario-specific mapping to cognitive debt, to be tested rather than presumed: investigators must examine whether the attitude or following changes involved deficient meta-audit, impaired external calibration, and identifiable, insufficiently tracked costs, and must evaluate alternative explanations such as voluntary continued following or ordinary preference updating before deciding whether to register them as cognitive debt. Voluntary continued following alone does not rule out that classification. These two experiments thus establish that algorithmic interventions can alter persuasion outcomes, political attitudes, and following behavior; they do not by themselves establish impaired meta-audit or the formation of cognitive debt. They provide existence evidence for projection-side influence, while their mapping to cognitive debt must still be assigned falsifiable conditions under §7.2. Computing power did not conquer the weather, but it is conquering attention.

This asymmetry deserves a pause for examination. The difference lies not in "which is easier to conquer"—but in the different nature of the defense mechanisms of the two systems. Atmospheric chaos is intrinsic and inalienable—no one can "draw a vacuum" on the weather. The defenses of the human cognitive system, by contrast, depend on a structural friction that requires continuous work to sustain. Once that work is interrupted—whether the convenience of automation strips away operational friction or algorithmic agreeableness strips away affective friction—the portion of the difference between the projection and defense sides that is not effectively filtered sediments within the system. This is the cognitive debt that this chapter will define.

§0.2 Another Asymmetry Diagnosed by von Neumann in 1955

In examining this cognitive asymmetry, we find that von Neumann, in 1955, had already diagnosed a formally similar asymmetry—only occurring on a different plane.

In "Can We Survive Technology?", he wrote:

> In the first half of this century the accelerating industrial revolution encountered an absolute limitation—not on technological progress as such but on an essential safety factor. This safety factor, which had permitted the industrial revolution to roll on from the mid-eighteenth to the early twentieth century, was essentially a matter of geographical and political Lebensraum: an ever broader geographical scope for technological activities, combined with an ever broader political integration of the world. Within this expanding framework it was possible to accommodate the major tensions created by technological progress. (von Neumann, 1955)

In the epigraph to this article, he judged:

> For the kind of explosiveness that man will be able to contrive by 1980, the globe is dangerously small, its political units dangerously unstable. (von Neumann, 1955)

What worried von Neumann was this: technological progress possesses an inherent global expansiveness—its radius of influence continuously swells until it covers the entire planet. Yet the vessel for absorbing this shock—the geographical boundaries of states, political structures, and the speed of institutional response—possesses a natural locality and sluggishness. In previous centuries, the problems created by the industrial revolution could be absorbed by expanding geographical and political units. But when the radius of technological influence reached the physical boundary of the earth, this safety mechanism of "expanding the vessel outward" failed.

His final judgment is sober and precise:

> For progress there is no cure. Any attempt to find automatically safe channels for the present explosive variety of progress must lead to frustration. The only safety possible is relative, and it lies in an intelligent exercise of day-to-day judgment. (von Neumann, 1955)

In an earlier passage, he pointed out:

> What safeguard remains? Apparently only day-to-day — or perhaps year-to-year — opportunistic measures, a long sequence of small, correct decisions. (von Neumann, 1955)

§0.3 The Formal Isomorphism of the Two Asymmetries

Von Neumann discussed the asymmetry between the geo-political vessel and the force of technological expansion. This chapter focuses on the asymmetry between the cognitive meta-audit vessel and the force of capital-technological projection. The two are not the same problem, but they share a formal characteristic:

The radius of influence of an expansionary force exceeds the scale of the carrying vessel; the original buffering mechanism fails; the difference is deposited directly within the system.

In von Neumann's diagnosis: the radius of technological influence covered the entire planet, and the geographical boundaries of political units could no longer expand to buffer the shock. The difference precipitated as international tension, nuclear standoff, ecological crisis.

In the framework of this chapter: the radius of influence of the capital-technological projection force is penetrating the cognitive boundaries of individuals, organizations, and states, while the cognitive defense mechanisms that absorb its shock—dependent on the continuous work of friction and meta-audit—are being systematically dissolved by the same technological logic. The difference precipitates as cognitive debt: a continuously accumulating stock left unaudited at the meta-level and invisible before it manifests.

This chapter does not claim that von Neumann foresaw the problem of cognitive debt—in the framework translation of this book, what he discussed was the geopolitical risk generated by the acceleration of the technological clock and the lag of the governance clock. But the formal structure he diagnosed—the asymmetry between an expansionary force and a local vessel—reappears in the cognitive domain in a form he did not discuss but which is formally recognizable. In *The Computer and the Brain* (1958), he also analyzed statistical notation in neural message systems and the trade-off between lower arithmetical precision and higher logical reliability. This book uses that difference as an engineering-side reference for understanding human cognitive architecture and goes on to propose a residual interpretation of cognitive debt. The latter step is this book's theoretical extension, not a

result already proved by von Neumann. Nor did he explicitly connect the asymmetry of the geo-political vessel with differences in neural notation in these two works.

§0.4 The Task of This Chapter

Chapter 1 provided the epistemological ground: within the Clock Room thought experiment, even billions of clocks are not time, readings are not reality, and mapping is systematically lossy. But Chapter 1 did not answer an operational question: when one is compelled to use clocks, how does one track the gap between multiple clocks and the true time? The task of this chapter is to construct this conceptual-analytic tool: a compulsory bookkeeping discipline that leaves the difference between the projection side and the defense side nowhere to hide. The difference must be entered into the ledger. Omissions must trigger alarms.

This bookkeeping structure is MCAF.

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§1 The Cognitive Extension of Double-Entry Bookkeeping: From Finance to Information

The MCAF framework inherits the core power of double-entry bookkeeping—not the bookkeeping technique itself, but the compulsory analytic discipline of "omission triggers alarm"—and extends it from the financial domain to the cognitive domain.

In 1494, Pacioli systematized double-entry bookkeeping in *Summa de arithmetica, geometria, proportioni et proportionalita* (Pacioli, 1494). Its core identity "Assets = Liabilities + Equity" established a strict financial discipline. Mattessich axiomatized the double-entry structure into a set-theoretic constraint system (Mattessich, 1964) and separately examined accounting from the problem of representation (Mattessich, 1987), revealing the core power of accounting—placing things of ontologically different natures (land, inventory, cash) into the same closed loop. The MCAF framework establishes "omission triggers alarm" as its own analytic discipline—an extension of the correspondence-recording idea of double-entry bookkeeping, not a detection capability that double-entry bookkeeping automatically realizes: omitting both sides of a transaction or misclassifying accounts leaves the trial balance intact. Macintosh and Quattrone independently confirmed this judgment from the perspective of organizational sociology: management accounting is not merely a measurement technique but a social device that renders organizational behavior visible and accountable (Macintosh & Quattrone, 2010).

The MCAF framework extends this technical discipline to the cognitive domain, fully inheriting three forms of theoretical self-awareness from modern accounting. **Teleological accountability**: Mattessich argued that the ultimate rationale for accounting is accountability encompassing ethical and social concerns (Mattessich, 2008)—accordingly, the core question of the MCAF framework is not "whether information transmission is efficient," but "who bears the structural friction generated by information projection and whether it is recorded in the books." **Functional social control**: Mattessich emphasized that accounting substantively concerns the performance of a social cybernetic function (Mattessich, 2015)—the MCAF framework models the projection-defense dynamic as a cybernetic problem, inheriting this disciplinary intuition. **Ontological commitment to social reality**: in the face of the anti-realist challenge that "income does not exist in the real world," Mattessich systematically argued that concepts such as income and debt belong to genuine social reality (Mattessich, 2014)—the core concept of the MCAF framework, cognitive debt, is positioned within this framework at the same ontological layer: an emergent property of the socio-cultural reality layer, not a rhetorical metaphor.

Precise Relationship to Double-Entry Bookkeeping. What the MCAF framework inherits from double-entry bookkeeping is the compulsory questioning discipline ("omission triggers alarm"), not its dual-entry bookkeeping structure ("every debit has its credit"). Strict double-entry bookkeeping requires the corresponding effects of a transaction to be entered in two or more accounts while keeping debits and credits in balance; the paired changes may occur between assets, liabilities, and equity, or within the same class of accounts—purchasing equipment with cash is an increase and a decrease within assets. The MCAF framework deliberately tracks only the liability side—the sedimentation of cognitive debt. This is because, within the current technological system, the benefits on the asset side (convenience of free services, increased efficiency of information access) are made immediately explicit by algorithms: a person instantly feels the benefit of a recommendation algorithm. The costs on the liability side (fragmentation of attention, degradation of judgment, erosion of institutional trust) are structurally concealed: a person does not feel what they are losing. The MCAF framework is a **selective tracking tool** designed to counter this asymmetrical concealment—it only illuminates the liability side, just as photographic developer reveals the latent image on a negative. It is not a complete cognitive balance sheet, but an asymmetrical discipline of debt tracking. The power of the framework comes precisely from this asymmetry: in an environment where the projection side is excessively illuminated and the defense side systematically obscured, unilaterally tracking the liability side is corrective, not biased. Here, "unilateral tracking" means that the output focuses on cognitive debt; to identify the source of debt increments, the analysis must still record changes on both the projection and defense sides.

Terminological Convention. This framework borrows the symbiotic classification system from ecology (de Bary, 1879). Licklider (1960) first introduced this concept into the human-machine relationship in "Man-Computer Symbiosis." This framework uses "symbiont" as a neutral umbrella term to refer to an external projection system that establishes an information coupling with the host; when, **in the empirical analysis of a specific scenario**, the cognitive debt output of that system is judged to systematically exceed the host's processing threshold, it **may, within that analytical context**, be labeled as a "parasitic party." This label is an analytical conclusion, not a presupposition of the framework.

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§1a The Applicable Boundary of the Framework: Minimal Counterexample and Minimal Delimitation

Minimal Counterexample. Not all technological enhancements deposit high cognitive debt. A hammer, a mechanical tape measure, a non-smart refrigerator, a traditional washing machine—these tools (**in their non-smart forms**) certainly also reorganize action and can fail, but they usually possess four characteristics: **clear functional boundaries** (a hammer is for driving nails), **directly perceivable failure consequences** (a refrigerator not cooling is immediately known), **external and accessible calibration mechanisms** (a tape measure can be checked against a national metrological standard), and the **existence of mature compensation-loop institutions such as return, repair, warranty, and recall**—if broken they can be fixed, defective ones returned; responsibility relies less on narrative transfer to the user, and also less on version updates to quietly conceal malfunctions or failures. In the language of the Clock Room thought experiment, these tools are clocks with clear graduations, provided with windows for calibration, and accompanied by return/replacement mechanisms. They are also clocks—and not time—but their narrative leverage ratio is low, their failure is visible, and the compensation loop is closed.

What this framework applies to is that class of capital-technology complexes that, while empowering, continuously expand their narrative boundaries, compress conditions for external calibration, and systematically transfer the responsibility for failure to the user—their narrative leverage ratio (§5.1 will formally define this) is much greater than 1, and their legitimacy base is in a state of stagnation or endogenous erosion.

Minimal Delimitation. This hereby clarifies the priority applicable boundary of the MCAF framework. Any tool whose functional boundary is highly clear, whose failure is highly visible, whose external calibration mechanism is stable, which primarily amplifies pre-existing human actions without systematically substituting human judgment and value-ordering, and which has relatively mature return, repair, and compensation mechanisms, is not a priority analytical object of this framework. Issues such as cost, efficiency, and labor substitution for such objects can of course still be discussed, but they need not be analyzed primarily in terms of "cognitive debt" or "narrative leverage ratio." The priority objects of the MCAF framework are those systems that simultaneously satisfy the following conditions: narrative boundaries are continuously extrapolated, external calibration is systematically compressed, failure costs are absorbed by users or the public rather than the projecting party, and the compensation loop is absent or dysfunctional.

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§2 The Meta-Formula and the Six Variables

§2.0 The Meta-Formula

The MCAF framework establishes the following meta-formula as the core bookkeeping template:

$$\Delta D_c = \underbrace{C \cdot \Lambda \cdot \phi}_{\text{Single Projection Load}} - \underbrace{A \cdot E_c \cdot \|L\|}_{\text{Single Defense Capacity}}$$

How to Read the Multiplication and Subtraction Signs. The multiplication and subtraction signs in the meta-formula are not instructions for arithmetic operations, but symbolic encodings of structural relations.

The multiplication sign encodes joint aggregation among factors on the same side: holding the other factors fixed, an increase in any one factor changes the product proportionally; if any key factor falls completely to zero, the product also becomes zero. These algebraic properties hold on both the projection and defense sides. The asymmetry between the two sides does not arise from multiplication alone, but from their different growth and maintenance conditions: unlocked stock (C), the media conductivity coefficient (Λ), and narrative load (ϕ) can each expand under capital and technological competition; meta-audit capacity (A), the human biological substrate (E_c), and the legitimacy base ($\|L\|$) must all remain operative, while being constrained respectively by continuous work, biological limits, and institutional legitimacy. The product form records joint effects and key-factor failure; it does not assign different algebraic logics to the two sides.

The subtraction sign encodes the net direction of two heterogeneous forces. The projection side and the defense side are ontologically incommensurable (§6 Postulate 2): high-frequency discrete information flows cannot be losslessly processed by the human continuous-statistical cognitive substrate. The minus sign should not be read as the arithmetic subtraction of commensurable quantities. Yet the collision of the two systematically produces a net direction—when the projection load exceeds defense capacity, ΔD_c is positive, and cognitive debt accumulates; when defense capacity exceeds the projection load, ΔD_c is negative, and debt is reduced.

A defense surplus is registered as a negative debt increment only when it is actually applied to identifying, correcting, or compensating existing debt; where no repayable debt exists, the surplus itself is not booked as negative debt stock. This net direction is ordinal (indicating increase or decrease), not cardinal (providing no commensurable quantitative difference). The meta-formula defines cognitive debt accordingly: ΔD_c is the ordinal residual precipitated after the collision of two heterogeneous structures.

The Hierarchy of the Six Variables. The six variables bear different levels of analytical function in the framework, and the reader should grasp their primary-secondary distinctions before entering the definitions of individual variables.

- **Anchor Variable**: The human biological substrate (E_c)—approximated as a constant on the time scale of civilizational history, providing the baseline reference for the entire temporal structure. The dynamical significance of all other variables is defined relative to E_c.
- **Change Variables**: The three projection-side factors (unlocked stock C, media conductivity coefficient Λ, narrative load ϕ) and the defense side's meta-audit capacity (A) —their states of change directly drive the increase or decrease of cognitive debt.
- **Carrying Variable**: The legitimacy base ($\|L\|$)—its primary analytical function is to set the critical boundary of the system's carrying capacity; its changes likewise alter defense capacity and affect debt increments. $\|L\|$ takes the minimum among its sub-items (the barrel effect: the system's carrying capacity is determined by the weakest link).

The definitions are unfolded variable by variable below, but the six variables do not bear equal explanatory weight in the framework.

The Mathematical Identity of the Meta-Formula. The meta-formula is an interdisciplinary chart of accounts, not a solvable dynamical system. Its mathematical identity is not an equation—it is a classificatory diagnostic discipline. The mathematical types of the variables (scalar, vector, matrix) are deliberately left undeclared at the level of the meta-formula; type declarations belong to the in-scenario projection mapping (§7.2, Level 2) and are determined by the specific scenario.

The meta-formula specifies three structural relations in advance: product-form aggregation on the projection side, product-form aggregation on the defense side, and a minimum constraint for the legitimacy base. At this level, they are structural encodings analogous to dimensional analysis, not commitments to directly computable numerical operations. Only after a scenario declares its variable types, scales, and aggregation operations can the strict forms of joint amplification, key-factor failure, and minimum constraint be determined.

Scope of Discussion on the Product Structure. The above discussion concerning the product structure applies to the aggregation relationships among the three variables within the projection side and among the three variables within the defense side, not to the commensurability between the projection side and the defense side. The ontological incommensurability between the two sides (§6 Postulate 2) does not cancel the legitimacy of the product structure within their respective sides. The product structure and the incommensurability declaration operate at different levels of relation: the former governs the manner of organization among variables within the same side; the latter governs the manner of comparison between the two sides.

The Usage Discipline of the Meta-Formula. The power of the meta-formula lies not in calculation—it lies in compulsory questioning: whenever the analyst describes a change on the projection side (such as technological

acceleration, increase in information flow density), the corresponding state of the defense side must simultaneously be recorded. An empty entry on either side is itself a diagnostic signal: the chart of accounts on which the analysis relies is incomplete.

Measurement-Theoretic Anchoring of the Product Structure. The formal structure of the meta-formula—"a small number of basic variables + monomial product aggregation"—has precedent for anchoring in classical measurement theory. Krantz et al. (1971), in Chapter 10 of *Foundations of Measurement*, proved that a system composed of "basic dimensions" and "multiplicative combination rules" is equivalent to a finite-dimensional rational multiplicative vector space (§10.2, Theorems 1–2), in which all secondary dimensions can be uniquely expressed as monomials of the basic dimensions; under this framework, any "dimensionally invariant" law must be rewritable as a relationship among dimensionless ratios (§10.3, Pi Theorem, whose initial form traces back to Buckingham 1914). This precedent has its own boundaries: the chapter's axiom system is a descriptive regimentation of the structure of physical quantities, not an argument for that structure's necessity; finite dimensionality is a declared assumption rather than something the theorems confer; and the book explicitly reserves for the nonphysical sciences a third road—arriving at lawful formulations whose character differs from the dimensionally invariant equations of physics.

The core claim of the meta-formula is not that "cognitive quantities have physical dimensions." This framework accordingly stipulates its own analytical form: organizing cross-scenario diagnosis through a small number of basic variables and monomial product aggregation, and requiring a scenario that adopts numerical scales to specify whether its diagnostic propositions preserve their shape under unit transformations. This is MCAF's own choice of analytical form, not a conclusion derived from measurement theory that every cognitive-social diagnosis must adopt monomial aggregation—the source explicitly reserves a third road for the nonphysical sciences (see the preceding paragraph). The six variables and product structure thereby constitute an analogical transplantation of this measurement-theoretic tradition into the context of cognitive debt—neither a usurpation of physical formulas nor a purely metaphorical decoration. For the scope and limits of its legitimacy, see §7 for in-scenario projection mapping and Chapters 9–16 (AI scenarios) and 17–22 (financial scenarios) for scenario calibration.

§2.1 Unlocked Stock (C)

Unlocked stock is ordered structure that has been released from sequestration and made available for mobilization by the projecting party. Its accumulation history documents how the resource was formed, but does not by itself determine the potential for narrative inflation; available scale, content quality, accessibility, and the concrete mode of use must each be booked separately.

The absolute equivalent of low-entropy bodies and ordered structures mobilized by the projecting party. "Unsealing" emphasizes that these resources are not created—they are released from some previously sequestered state. Fossil fuels sequestered hundreds of millions of years of solar radiation energy. Training data corpora sequestered five thousand years of human written knowledge. Genomic information sequestered billions of years of evolutionary trial and error.

The four categories of ordered structure—material, informational, biological, and attentional—may have different accumulation and unsealing time-scales, and particular acts of unsealing may be irreversible. The scale and quality of available ordered structures constrain the possible boundary of narrative load; accumulation time

alone is not a cross-scenario monotonic conversion scale.

§2.2 Media Conductivity Coefficient (Λ)

The media conductivity coefficient is the efficiency with which information reaches the host—not an exogenous constant, but a technological variable shaped by power-holding actors, possessing a macro-meso-micro three-level nesting.

Three-level nested structure: The macro-infrastructure layer determines the physical reachability of information (navigation → fiber optics → satellite internet). The meso-distribution architecture layer determines the actual allocation strategy of information. The shift from open protocols to algorithmic recommendation engines means that control over information distribution moves from decentralized arrangements to platform objectives. The micro-contagion dynamics layer describes the diffusion and attenuation characteristics of narratives among individuals.

The three layers co-evolutionarily amplify each other. The inflation of the media conductivity coefficient is not growth in a single dimension, but the co-evolution of a three-level structure. This synergistic amplification within the three levels and the product-form aggregation described in §2.0 are two different relations: the former describes positive feedback among levels within the same variable; the latter jointly records the states of different variables in the projection load. Both bear on the media conductivity coefficient (Λ): positive feedback within the three levels affects changes in Λ itself, while product-form aggregation incorporates those changes together with the states of unlocked stock (C) and narrative load (ϕ).

§2.3 Narrative Load (ϕ)

Narrative load is the **narrative packaging and information asymmetry** projected by the projecting party in pursuit of local optima. Borrowing a metaphor from calculus: its **first-order change** (the turnover of specific narratives) is visible and auditable; its **second-order change** (the accelerated cementing of underlying narrative frameworks) weakens the contrast conditions required for auditing because it uniformly blankets the information environment.

"Local optimum" means optimization for the projecting party's own interests (more engagement, longer duration, higher conversion rates), not optimization for the system as a whole (including user welfare). "Narrative" broadly refers to any signal structure that causes the receiver's behavior to deviate from the choice they would make under perfect information—the nested complexity of financial products, the personalized precision of algorithmic recommendations, the opaque outputs of AI models, are all different carriers of narrative load. Narrative load measures the total amount of narrative actually released by the projecting party; the deviation-from-perfect-information characterization in the preceding sentence describes a property of its typical carriers, not a defining criterion of the total.

The temporal behavior of narrative load is expressed through a calculus metaphor involving two orders. First-order change corresponds to shifts in specific narratives and is generally easier to observe and audit. Second-order change ($d^2\phi/dt^2 > 0$) corresponds to underlying assumptions becoming industry defaults or even civilizational defaults at an accelerating rate. Its difficulty of audit does not follow from derivative order itself, but from the uniform coverage of the information environment, which weakens the contrast conditions needed for audit. The detailed discussion appears in §3 (Variable Time Signatures); applications of second-order change

in narrative load to AI and financial scenarios appear in Chapters 9–16 and 17–22, respectively.

§2.4 Meta-Audit Capacity (A)

Meta-audit capacity is the continuous work required by a subject to maintain an independent orthogonal coordinate system, separate from the narrative closed-loop of the projecting party—without work, there is no coordinate system; without a coordinate system, categorical judgment is impossible.

Core Definition. Meta-audit capacity is not computational depth (how far a chain of calculation can proceed)—it is the existence of a coordinate system (whether one can see that two narratives belong to the same projection structure type). Without continuous work, there is no independent coordinate system; without an independent coordinate system, categorical judgment is impossible. The essence of meta-audit is not "calibrating a clock with a clock" (ordinary auditing—Chapter 1 §1.3 has already argued this is merely a closed loop), but boring a hole in the closure of information: introducing an external reference not belonging to the current narrative closed loop, giving the difference an opportunity to be exposed.

Research on complex-estimate auditing provides an empirical reference for the decay of meta-audit capacity within an established process. Drawing on interviews with 24 partners and senior managers from six large audit firms and on inspection reports issued by the Public Company Accounting Oversight Board (PCAOB), Griffith, Hammersley, and Kadous (2015) find that auditors overwhelmingly reported testing management's process for generating complex estimates rather than using an approach that relies less on management's assertions. Professional standards and firm policies emphasize verifying components of management's model rather than critically evaluating the reasonableness of the estimate. Mapped into MCAF as a scenario-specific proposition: when the model supplied by management becomes the boundary of subsequent verification, external evidence may lose its role as an independent reference, and meta-audit capacity is weakened within the existing process.

Ontological Anchoring. Meta-audit capacity is ontologically anchored in an **anthropological constant**: the continuous energy output required for a subject to maintain the structural equilibrium of their core social-emotional bonds (intimate relationships and role networks). The complete unfolding of this anchoring—based on Francis L. K. Hsu's Psychosocial Homeostasis (PSH) theory—appears in Chapter 5. Hsu pointed out that human society links its members with two things: roles (functional, measurable by money) and affect (intimate feelings that cannot be bought or sold) (Hsu, 1971, 1983). Meta-audit capacity is accordingly divided into two orthogonal dimensions: **Role audit (A_{inst})** deals with tradeable, replaceable functional relationships—whether efficiency is reasonable, whether procedures have been hijacked. **Affect audit (A_{value})** deals with non-tradeable, irreplaceable intimate bonds—what is worth auditing, whether the independent coordinate system has been eroded by narrative. The two are orthogonal: the precise operation of role audit cannot compensate for the decay of affect audit. Here, orthogonality is a hypothesis of functional non-substitutability, not a claim that the two empirical variables are statistically independent; role and affect may overlap within the same relationship (Chapter 5, §2.2). Hsu's criterion: "a society in which no one will do anything except for money is in serious trouble" (Hsu, 1983).

Hierarchical Structure. Meta-audit capacity uses a three-level product-form aggregation: $A = A_{macro} \cdot A_{meso} \cdot A_{micro}$, corresponding respectively to macro-level institutional self-reflection, meso-level organizational/sectoral auditing, and micro-level daily extension of innate knowledge. If any level is completely absent, overall meta-audit capacity fails; if a level merely weakens while remaining nonzero, the other levels may pro-

vide limited compensation under scenario-specific constraints. The role/affect distinction spans all levels. The complete unfolding of the eight layers of PSH and their detailed mapping to MCAF variables appears in Chapter 5.

§2.5 The Human Biological Substrate (E_c)

The human biological substrate is the very manner in which human beings, as embodied organisms, process information from the real world. It is not a parameter inside the brain, but the entire embodied architecture through which the host interacts with the environment—how the senses receive, how the nerves organize, how consciousness processes. This architecture is constant on the time-scale of civilizational history; it is the anchor of the entire temporal structure.

One Architecture, Two Operating States, Two Levels of Analysis. The human biological substrate is the same neural matrix operating at different task depths; it is not an assembly of two separate systems. Neural message encoding and conscious symbolic reasoning belong to different levels of analysis.

At the shallow level, the system operates in a wide-field scanning state: sensory systems receive environmental information in parallel at an extremely high rate—approximately 10^9 bits/s (Zheng & Meister, 2025)—and detect anomalies through mid-level cues such as the rapid capture of the gist of a whole scene (Oliva & Torralba, 2006), contour features, stylistic deviations, and violation signals. Certain natural-scene categorization tasks can be performed in the near absence of conscious attention (Li et al., 2002); whether other forms of anomaly detection, violation signaling, and threat recognition are equally attention-light must be tested task by task. Von Neumann's analysis of the neural message system supplies a constraint at the underlying level: meaning is carried by statistical properties of pulse trains, such as frequency, rather than by the precise positions of symbols. The cost of this statistical encoding is low arithmetic precision—approximately two to three decimal digits—while its benefit is high logical reliability: losing a few pulses does not completely distort the message's meaning (von Neumann, 1958, ch.15).

Once an anomaly is located, the same neural matrix shifts into deep operation: conscious, serialized attentional processing with extremely low information throughput—approximately 10 bits/s (Zheng & Meister, 2025)—and a working-memory capacity of approximately 4 ± 1 chunks (Cowan, 2001), but with extremely high precision. At the cognitive and representational level, this operating state supports conscious symbolic reasoning: precise logical chains transmit information, and errors at one step affect subsequent steps. This does not mean that the underlying neural message system switches from statistical encoding to the positional symbolic encoding of a digital computer. Von Neumann instead described conscious mathematics as a secondary language built on the nervous system's primary language (von Neumann, 1958, ch.16).

The two states are not "one good, one bad"—they are two operating points of the same architecture, corresponding to two different modes of cognitive activity and serving two different functions. The shallow level answers "where is something wrong", the deep level answers "why is it wrong." The configuration granted by evolution is: the shallow level is nearly free; the deep level is exceedingly expensive.

The Scope of the Evolutionary Explanation. Von Neumann's *The Computer and the Brain* (1958) uses neuronal physical parameters to argue for limited logical and arithmetical depth and for the architectural significance of massive parallelism. This engineering argument does not by itself prove a particular evolutionary history. In this book's functional interpretation, shallow wide-field scanning and deep serial reasoning represent different

operating configurations across information-acquisition cost, response speed, and metabolic expenditure. Their evolutionary formation must be established through evidence about ecological niches, metabolic trade-offs, and cross-species comparison, rather than foreclosed by the assertion that a "deep and narrow" machine cannot survive.

Von Neumann further identified the intrinsic reason logical depth is constrained: in a statistical encoding system, the error of each computational step not only accumulates but is also **amplified** by subsequent steps. He took the retina as an example—the reorganization of the visual image effected on the retina, or more precisely at the point of entry of the optic nerve, passes through only three consecutive logical steps (three successive synapses). The extremely small logical depth is not because the brain "cannot compute," but because the error propagation mechanism of the statistical system determines that the logical chain cannot be too deep (von Neumann, 1958, ch.15–16).

This book accordingly treats shallow wide-field scanning and deep serial reasoning as two analytical operating modes, without presuming either family of tasks to be the sole evolutionary default in every setting. Which mode a particular task invokes, and how much attention and metabolic cost it incurs, must be determined by task-specific evidence.

The Position of E_c in the Meta-Formula. E_c is the anchor variable among the three defense-side variables ($A \cdot E_c \cdot \|L\|$). The other five variables are all changing; E_c is approximately invariant. It is the only variable in the meta-formula that can be treated as a constant on the time-scale of civilizational history.

What E_c Marks Is Not Just Speed, but an Entire Information-Processing Architecture. Von Neumann wrote in the unfinished final chapter of *The Computer and the Brain* that the logic and mathematics of the central nervous system "must structurally be essentially different from those languages to which our common experience refers" (von Neumann, 1958, ch.16). At the underlying level, the neural message system transmits information statistically; the mathematics and logic used consciously are secondary symbolic languages built on that primary language. Digital computers rely on precise positional encoding and deeper serial operations. The difference is therefore not that the human brain adopts computer-style symbolic encoding in deep operation, but that a statistical neural substrate bears secondary symbolic reasoning in a manner fundamentally different from a digital computer.

Numerical Precision and Qualitative Constraints. This book's use of E_c relies on only two properties: (1) sensory-input rates and the information throughput of conscious behavior differ by roughly eight orders of magnitude, while shallow wide-field scanning and deep serialized reasoning serve different cognitive functions; and (2) this embodied architecture is approximately constant on the time-scale of civilizational history—the rhythm of biological evolution, measured in millions of years, is far slower than the rhythm of technological evolution, measured in years to months. Zheng and Meister (2025) provide the empirical basis for the difference in information rates. Von Neumann (1958) provides an architectural argument for the statistical character and limited logical depth of neural message systems. This section therefore distinguishes two modes of cognitive activity; it does not claim that the underlying neural code switches with the task. Even if future measurements revise the estimate of conscious information throughput to 20 or 5 bits/s, these two qualitative properties still hold. If future neuro-enhancement technologies alter the upper bound of conscious human information processing, this anchor point will need revision—but at the current stage of civilization, the approximation remains robust.

§2.6 The Legitimacy Base ($\|L\|$)

The legitimacy base is constrained by whichever component first reaches its failure boundary—the level axis and the category axis must be treated separately, and excess accumulation in one dimension cannot automatically compensate for the absence of another.

The underlying consensus, law, and physical-isolation entities on which the host depends form a three-level × three-category matrix. **Level dimension**: macro (sovereignty and infrastructure shells), meso (organizational norms and recourse buffers), micro (physical and biological boundaries—the last load-bearing wall after all institutional failures). **Category dimension** (Zhao Dingxin's three-dimensional analysis): performance legitimacy (L_{perf}) is anchored in visible growth dividends; ideological legitimacy (L_{ideo}) is based on historically sedimented identification and belief and evolves slowly; procedural legitimacy (L_{proc}) provides the rules for succession and lawful procedures of power, constituting the resilience of the institutional layer against short-term fluctuations (Zhao Dingxin, 2016).

Aggregation proceeds in two stages. Within each level, performance, ideological, and procedural legitimacy first form that level's composite value according to falsifiable rules defined for the specific scenario; the meta-formula does not prescribe this within-level function. The overall legitimacy base then takes the minimum across levels: $\|L\| = \min(\|L_{macro}\|, \|L_{meso}\|, \|L_{micro}\|)$. Overall carrying capacity is therefore constrained by the currently weakest level; relations among the three categories belong to within-level empirical mapping and must not be conflated with minimum aggregation across levels.

Under normal conditions, limited compensation may occur among the three categories of legitimacy—very high performance legitimacy can temporarily mask the absence of procedural legitimacy, and high growth often masks institutional defects. But this within-level empirical relation is brittle: once declining performance or an exogenous shock causes a level's composite value to reach its failure boundary first, the compensation fails. The timing of threshold fracture depends on current levels, scenario-specific thresholds, and evolutionary paths; it is not determined by the fastest rate of decay alone. MCAF, as a debt-tracking mechanism, focuses on this critical brittleness.

§2.7 The Asymmetry of Growth and Maintenance Conditions

The projection and defense sides use the same product-form aggregation, but their variables differ in growth conditions, constraints, and the distribution of control.

Holding other factors fixed, an increase in any projection-side variable raises the total load; capital and technological competition can separately accelerate unlocked stock, the media conductivity coefficient, or narrative load. Defense capacity, by contrast, requires meta-audit capacity, the human biological substrate, and the legitimacy base to remain simultaneously operative, while these factors are constrained respectively by continuous work, biological limits, and institutional legitimacy. The asymmetry discussed here is therefore an asymmetry of empirical mechanisms and constraint structures, not two different logics inherent in multiplication.

Complete absence must be distinguished from nonzero decay. If any key factor falls completely to zero, the corresponding product on either side also becomes zero; if a factor merely decays while remaining nonzero, changes in other factors may still provide limited compensation. The hierarchical aggregation of meta-audit capacity and the minimum constraint on the legitimacy base further specify the defense side's modes of vulnerability: institutional auditing may compensate for insufficient individual auditing within limits, but it cannot

replace a completely absent key factor; the legitimacy base remains constrained by the currently weakest level.

§2.8 The Action Criterion for the Positions of Bearers

The meta-formula specifies the structural relationship between the projection side and the defense side, but no real-world actor corresponds permanently to either side. Real-world markets are institutions jointly constituted by laws, practices, and norms, and governments, firms, and individuals all participate in constructing and governing them (Vogel, 2018). The identity of a state, organization, or individual does not by itself determine its position in MCAF; classification must turn on how its concrete actions actually affect the six variables.

In practice, both projection and defense are jointly produced by states, organizations, individuals, and other actors, but these actors do not correspond one-to-one to the macro, meso, and micro levels: platform firms operate in daily practice at the meso-distribution architecture layer, yet their data centers and cloud networks are components of the macro-infrastructure layer; state intervention need not remain at the macro level either—a ban targeting a single product form acts directly on a micro-level target. Here, level refers to the reach of an action, not to the identity classification of macro-, meso-, and micro-level bearers of meta-audit capacity. Levels are determined by the scope on which an action operates, not by the category of the actor. In a given setting, state licensing, infrastructure construction, or expanded procurement may increase unlocked stock (C) or the media conductivity coefficient (Λ); independent auditing, disclosure, and redress mechanisms may instead increase meta-audit capacity (A) or stabilize the legitimacy base ($\|L\|$). Firms, platforms, rating agencies, industry associations, and professional communities may increase the media conductivity coefficient (Λ) and narrative load (ϕ) through distribution architectures and narrative packaging, or assume meso-level defensive functions through independent standards, monitoring, and accountability mechanisms. Individuals who cede data and attention increase the resources available to the projection side; cross-source verification and the maintenance of independent judgment constitute the actual work of micro-level meta-audit.

Accordingly, MCAF does not substitute the distinctions between "government and market," "public and private," or "organization and individual" for the distinction between the projection and defense sides. Moreover, in MCAF analysis, treating these dichotomies as fixed ontological boundaries produces a second-order narrative that itself requires audit: describing regulation as "intervention" in a market that "should be free" presupposes a self-regulating market that never existed as the baseline rather than providing a neutral analytical starting point. Each scenario mapping must specify which actions alter unlocked stock, the media conductivity coefficient, and narrative load; which actions sustain meta-audit capacity and the legitimacy base; where the human biological substrate imposes constraints; and whether the same actor operates on both sides. On this basis, MCAF adds the analytical step of recording these actions separately according to how they affect projection load and defense capacity. It must be emphasized that for the state, "operating on both sides simultaneously" is not a contingent coexistence but a structural drive: the state often maximizes performance legitimacy by authorizing the inflation of the projection side, while at the same time overdrawing the long-term carrying capacity of the legitimacy base. Chapter 5 further examines, through the Psychosocial Homeostasis (PSH) model, why states in different civilizations develop different default configurations, and why this paradox runs deeper in some polities than in others.

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§3 Temporal Structure: Three Clocks and the Time Signatures of Variables

The six variables each carry a time signature, converging into three macroscopic clocks: the techno-capital clock (often months; scenario-calibrated), the governance clock (often years; scenario-calibrated), and the debt clock (scenario-calibrated; decades to generations in some paradigmatic cases in this book, not a universal value).

§3.1 Time Signatures of the Variables

The six variables of the meta-formula each carry different temporal properties. These properties are not ancillary labels—they constrain the behavior of the variables in continuous time, thereby influencing when the velocity difference between the projection side and the defense side appears and at what rate it widens.

The temporal properties of unlocked stock must separately record accumulation history, available stock, and stock growth. Oil sequestered hundreds of millions of years of solar radiation, and large language models compress approximately five thousand years of human written corpora into hundreds of billions of parameters. These cases concern different formation histories, but they do not establish a ranking of narrative potential by accumulation time alone. Stock growth may be driven by capitalist competition—using AI infrastructure as an example, total global AI computing capacity has roughly doubled every 7 months since 2022 (You et al., 2026), and the training compute of notable AI models has roughly doubled every 6 months since 2010 (Rahman & Owen, 2024)—but stock growth and accumulation history are not the same account. In the specific scenario of training corpora, the finite stock of public human text can constrain continued expansion in dataset size. Villalobos et al. (2024) estimate that, if current trends continue, training datasets will approach the available stock of public human text between 2026 and 2032; the same study also discusses synthetic data, transfer learning, data-efficiency improvements, and non-public data as possible mitigations. This evidence therefore supports the conditional constraint of a particular finite resource, not a general law that the narrative potential of all four categories of ordered structure must decline after exhaustion. Other scenarios require separate calibration of stock, quality, and substitution mechanisms.

Accumulative Development Time and Diversity Time. The reason why the three variables on the projection side and the three variables on the defense side have completely different time signatures is rooted not in technical details, but in the difference in the nature of the human competition on which they rely. Starting from the structure of social power, Zhao Dingxin (2019) distinguished two fundamentally different temporalities.

Economic and military competition share two common features: clear criteria for winning and losing, and sustained selection pressure for organizational efficiency and productive capacity. These two features enable economic and military competition to drive the accumulative development of history—what Zhao Dingxin termed "accumulative development time." Accumulative development has a direction (increasing efficiency) and an accelerating mechanism (competitive selection pressure eliminates participants with slower growth rates), but it does not imply "progress"—it simultaneously brings environmental crises, global injustice, and weapons capable of annihilating humanity. The exponential inflation of the three projection-side variables (unlocked stock, media conductivity coefficient, narrative load) runs precisely on accumulative development time: the release of unlocked stock is driven by capitalist competition, the three-level nesting of the media conductivity coefficient is driven by technological races, and the first derivative of narrative load is driven by market selection pressure. They are the concrete unfolding of accumulative development time in the cognitive domain.

Zhao Dingxin's ideal-typical analysis offers a reference for some, not all, temporal properties of the defense side.

Ideal-typical ideological competition lacks clear victory criteria and is persuasive and highly pluralistic; the formation of values in meta-audit and the accumulation of ideological legitimacy may therefore be interpreted primarily through diversity time. Procedural legitimacy, political authorization, and institution-building are also shaped by political competition, organizational resources, and administrative procedures, and cannot be uniformly assigned to diversity time. The speed difference between the techno-capital and governance clocks is accordingly a structural proposition to be tested in concrete scenarios. Differences between value-consensus formation and economic-technological competition may supply part of the explanation, but they do not rule out political competition, organizational capacity, or institutional design; nor can governance delay be equated simply with institutional inefficiency.

The temporal property of the media conductivity coefficient is a three-level nested time-scale. §2.2 has already defined the three-level structure of the media conductivity coefficient (Λ), each level having its characteristic time-scale: the macro-infrastructure layer evolves with a period of decades; the meso-distribution architecture layer updates with a period of weeks to months; the micro-contagion dynamics layer fluctuates with a period of hours to days. The synergistic effect of the three levels makes the effective growth rate of the media conductivity coefficient faster than any single level—the expansion of infrastructure provides a larger operating space for algorithmic iteration; the optimization of algorithmic iteration provides more efficient amplifiers for contagion dynamics.

The temporal property of narrative load is that of a typical capital-technological clock. The rate of change of narrative load (ϕ) is driven by the pressure of capitalist competition. The first derivative of narrative load corresponds to observable narrative switching—such as the replacement of a certain advertisement or the adjustment of a platform policy—these changes, because they are visible, can be audited. The second derivative of narrative load ($d^2\phi/dt^2 > 0$) corresponds to the process by which underlying assumptions accelerate and cement into industry defaults and even civilizational defaults. When these accelerating processes uniformly blanket the information environment, they weaken the contrast conditions required for audit. When global users use the same set of models to obtain information, organize their thoughts, and assist their writing, the model's default framework becomes the user's default framework—not because the user has been persuaded, but because alternative frameworks have disappeared from the information environment. A fish cannot audit water. This difficulty of observation arises from uniform coverage and the loss of contrast conditions, not from derivative order itself. Second-order change is not an independent component of narrative load but a higher-order dynamic of the same quantity that can be observed through proxy indicators in specific scenarios. For the civilization-skewing effects of the second derivative of narrative load in AI scenarios (asymmetries in corpus structure, tokenizer efficiency, epistemological frameworks), see Chapters 9–16 in detail.

The temporal property of meta-audit capacity is a development-decay asymmetry. The construction of meta-audit capacity (A) depends on educational cycles (years to decades), institution-building (decades to generations), and social learning (extracting lessons from crises and institutionalizing them, years to decades). Meta-audit capacity can decline faster than it is built. Bainbridge's analysis of the ironies of automation points out that operational skills deteriorate through disuse and that effective retrieval of long-term knowledge depends on practice and feedback (Bainbridge, 1983). This supports the mechanism of deterioration but does not supply a uniform time constant for the decay of meta-audit capacity. Construction is slow, decay is rapid—this asymmetry makes the long-term effective value of meta-audit capacity lower than its peak level.

The behavior of meta-audit capacity at the individual and institutional levels is fundamentally different. Indi-

vidual micro-level meta-audit capacity is subject to the order-of-magnitude constraint of the human biological substrate of approximately 10 bits/s—each act of auditing occupies a portion of this magnitude, so individual audit output cannot exceed that order of magnitude. Institutional macro-level meta-audit capacity bypasses the processing limit of any single individual. Its pathway of action is not to match the projection side with greater defense capacity, but to modify projection-side variables directly through the exercise of power: prohibiting specific product designs truncates unlocked stock, limiting the nesting layers of financial products compresses narrative load, and the requirement for algorithmic transparency can directly reduce model opacity and the difficulty of independent verification within narrative load. An order of prohibition contains very little information—a few hundred words—but its compression effect on the projection-side product can be global. Institutional intervention can have effects far beyond the processing capacity of any individual; however, the timing of its implementation is constrained by the governance clock, the scope of its exercise by authorization from the legitimacy base, and the extent to which it can correct past effects by the sunk costs of unlocked stock.

The human biological substrate has no clock. The human biological substrate (E_c) is approximately constant on the time-scale of civilizational history. This is the anchor of the entire temporal structure.

The temporal property of the legitimacy base is slow accumulation—rapid collapse. The categories of the legitimacy base ($\|L\|$) have different time-scales: the institutional stock of performance legitimacy takes time to accumulate, but its current level of support can shift rapidly after a technological dividend or systemic failure; procedural legitimacy depends on transparency and rule stability, and each algorithmic update by the projecting party depreciates prior transparency claims; ideological legitimacy depends on consensus and has a generational time-scale. The moment of overall failure depends jointly on current levels, scenario-specific thresholds, and evolutionary paths; whichever level first reaches its failure boundary becomes the binding constraint.

§3.2 Definition of the Three Clocks

The time signatures of the six variables converge into three macroscopic clocks.

The MCAF framework denotes the techno-capital clock by τ_c, the governance clock by τ_g, and the debt clock by τ_d. The techno-capital clock marks the characteristic period over which the projection side changes significantly; the governance clock marks the period required for the defense side to complete an institutional response; and the debt clock marks the characteristic delay from the sedimentation of cognitive debt to its first manifestation. The asymmetry among these time-scales is the root of cognitive-debt dynamics.

The techno-capital clock (τ_c) is the characteristic period over which the product of the three projection-side variables changes significantly—the faster the product grows, the shorter the period. Competitive selection pressure drives this clock: participants whose projection-side product grows faster seize resources and narrative space, further shortening the next cycle of change. In the Clock Room thought experiment, the techno-capital clock marks the interval at which new clocks are manufactured and placed before the subject. Zhao Dingxin's "accumulative development time" gives this clock its sociological character: directional, cumulative, and productive of isomorphic pressure (Zhao Dingxin, 2019).

The governance clock (τ_g) is the full response cycle of the defense side acting on the projection side through institutional power—including the four sub-processes of identifying the problem, verifying the evidence, formulating the intervention, and implementing the intervention. The denominator of the governance clock is not the product of the individual's micro-level meta-audit capacity and the human biological substrate—it is the

product of the institutional-level macro meta-audit capacity and the legitimacy base. Institutional-level meta-audit capacity does not run through each individual's human biological substrate; thus, the governance clock is not anchored by this individual upper bound. In the language of the Clock Room thought experiment, the governance clock is the time required from "discovering that the clock may be wrong" to "taking action to calibrate or remove the clock."

The lower bound of the governance clock must be estimated separately for each institutional setting rather than stipulated as a uniform incompressible constant. Regulatory intervention requires both verifiable evidence and legitimate authorization. Crash-test data can support the effectiveness of automotive safety standards, and financial-crisis data can support the necessity and efficacy of bank capital-adequacy interventions; how these materials are converted into procedural or performance legitimacy must be further specified through authorization procedures, relevant publics' recognition of those procedures, and the provision of public goods; their presence alone cannot predetermine the result. Producing new evidence may take time, but invoking existing evidence, obtaining emergency authorization, and executing an administrative response have different cycles. Which components can be compressed, and at what evidentiary or legitimacy cost, must be specified item by item. Value formation and consensus-building in governance may be interpreted through diversity time, but political authorization and administrative execution are also shaped by political competition, organizational resources, and institutional design. The difference between the techno-capital and governance periods must therefore be calibrated in each scenario rather than predetermined by a single temporal ontology.

The debt clock (τ_d) is the characteristic delay from the sedimentation of cognitive debt to its first manifestation. Cognitive debt sediments at the moment of interaction—the consumer absorbs an unaudited narrative load when purchasing a product with a hidden safety defect; the retail investor incorporates noise as a signal into their decision when buying an attention-stock driven by emotion. The debt manifests at the moment of invocation—the car owner discovers the design flaw exposed when encountering an accident; the retail investor discovers they bought at the top when the price mean-reverts. The time interval between sedimentation and manifestation is the graduation of the debt clock. The length of the debt clock is not fixed—it depends on when the sedimented error is triggered. This uncertainty makes the debt clock the most unpredictable of the three clocks—the observer knows the debt is there but does not know when it will manifest. In the language of the Clock Room thought experiment, the debt clock is the time that elapses from when the subject accepts an uncalibrated reading until the falsity of this reading is exposed with some consequence in the real world.

§3.3 Structural Ordering

The ordering of the three clocks is a conditional diagnostic rather than a universal normal state $\tau_c \ll \tau_g \ll \tau_d$ for every scenario. Under the competitive conditions specified in this chapter, the techno-capital clock is generally the shortest; this is an empirical proposition to be tested in each setting. The relative positions of the governance and debt clocks are determined by actual response and manifestation: if governance responds before debt first manifests, then $\tau_g < \tau_d$; if debt manifests before the governance response is complete, the system enters a crisis state.

The techno-capital clock τ_c is generally the shortest because competitive selection pressure eliminates participants with slower growth rates—participants whose projection product grows faster seize resources and narrative space, while slower participants are eliminated or forced to follow. Locked-in infrastructure and continuing competition further shorten the next cycle of change.

The governance clock must separately record the time spent on identification, evidentiary verification, policy design, authorization, and implementation; problem identification cannot be presumed to be the most time-consuming component in every governance process. Skipping necessary evidence, public deliberation, or lawful procedures may damage the intervention's legitimacy and durability, but this cost does not entail a single incompressible lower bound for all governance.

The debt clock is difficult to predict because manifestation depends on specific trigger conditions that no single participant fully controls. Unpredictability does not entail the longest period. Institutional arrangements can alter trigger probabilities, timing, and propagation paths; the debt clock is therefore not wholly unresponsive to human decisions, but it is not fully governed by any single actor.

In the language of the Clock Room thought experiment, this conditional relation means that new clocks are usually pushed into the room faster than the subject can calibrate them. When the falsity of previously accepted uncalibrated readings will become exposed must be recorded against specific trigger conditions; the metaphor must not presume that exposure necessarily occurs after governance is complete.

The Contraction of the Present. Rosa (2013) provided an independent theoretical reference for the tension between the techno-capital clock and the governance clock from the perspective of critical theory. He identified the self-reinforcing circuit of the triple acceleration of modern society: technological acceleration (transportation, communication, production), the acceleration of social change (the shortening replacement cycles of cultural knowledge, social institutions, personal relationships), and the acceleration of the pace of life (despite technological progress supposedly increasing free time, the actual pace of life instead quickens). The three are mutually causal: technological acceleration drives the acceleration of social change; the acceleration of social change forces individuals to accelerate their life pace; and the acceleration of life pace creates demand for even faster technology.

Rosa's most important concept is the "contraction of the present"—the time window within which expectations based on past experience can reliably match the future is continuously shrinking. Translated into the language of the MCAF framework: the effective period of the last round of calibration is continuously shortening. When the techno-capital clock (τ_c) accelerates to a certain point, each response of the governance clock (τ_g) is already outdated before it is completed—it calibrates a projection-side state that no longer exists. This not only provides a reference for the ordering $\tau_c \ll \tau_g$ but also reveals a dynamic possibility: the gap may widen. The techno-capital clock (τ_c) is self-compressing—the next cycle is shorter than the last—while value formation and consensus-building within the governance clock (τ_g) may be constrained by diversity time, and the compressibility of evidence production, political authorization, and administrative execution must be calibrated in each institutional scenario. When the governance period compresses more slowly than the techno-capital period, their ratio (τ_g/τ_c) increases.

Rosa also identified a paradoxical symptom: "frenetic standstill"—a metaphor Rosa credits to "an inspired translation of Paul Virilio's inertie polaire"—people feel they must run harder, faster, yet do not feel they are actually going anywhere. In the language of the MCAF framework, this is the subjective experience after the cognitive Reynolds number enters the turbulent interval: the density of information arrival far exceeds information throughput; the individual is forced to abandon item-by-item auditing and resort to heuristics and meta-clocks—but the construction of the meta-clock itself is also accelerating; every meta-clock is outdated before it is even built. So the subject frenetically reads the readings, but the distance between the subject and

reality does not shrink at all.

The relative position of the governance clock (τ_g) and the debt clock (τ_d) **is not a fixed parameter**—it is a diagnostic variable of the system state.

The ordering $\tau_c \ll \tau_g \ll \tau_d$ above describes an ideal-typical cumulative scenario in which debt remains untriggered for an extended period, rather than defining "normalcy" across scenarios: in this scenario, debt has not yet manifested, and governance has time but lacks a legitimate basis for action—this is the first half of the Collingridge dilemma. When a trigger condition suddenly makes accumulated debt manifest, manifestation precedes the completion of the governance response. This does not rewrite the historical characteristic period τ_d after the trigger; rather, the remaining waiting time falls to zero. This is the second half of the Collingridge dilemma. Collingridge's original formulation reads:

> This is the dilemma of control. When change is easy, the need for it cannot be foreseen; when the need for change is apparent, change has become expensive, difficult and time consuming. (Collingridge, 1980)

The debt clock (τ_d) **is not a constant-speed clock**. Before a trigger condition appears, the specific moment of manifestation is unpredictable. When the trigger occurs, what falls to zero is the remaining waiting time from the present to manifestation, not the sedimentation period already elapsed. Direct loss may become manifest without a proximate signal, but a shortening verification window and deteriorating proxy-risk indicators may still appear in advance. The early-warning task in §7.5 concerns these proxy signals, not the precise prediction of the trigger moment.

While the competitive conditions specified in this chapter persist, the techno-capital clock (τ_c) is generally the shortest. Whether the system has entered crisis is determined not by rewriting the historical lengths of the three clocks, but by whether debt manifests before the governance response is complete. Under normal conditions, the governance predicament is **political**: there is time to govern, but no legitimacy basis for action. Under crisis conditions, the predicament is **physical**: the evidence has arrived, but the governance response remains incomplete. The former requires preventive judgment; the latter requires emergency speed.

§3.4 The Bifurcation at the Individual Level and the Institutional Level

If the projection side continues to grow while improvement on the defense side remains constrained, the individual cognitive Reynolds number faces long-term divergence pressure. Institutional-level overload pressure does not necessarily diverge: it depends on whether the governance response arrives before projection-side lock-in and whether it is completed before debt manifests.

Individual Level. In the denominator of the individual cognitive Reynolds number, the human biological substrate is constant, improvement in micro-level meta-audit capacity is constrained by that substrate, and improvement in the parts of the legitimacy base concerning value identification and ideological accumulation is constrained by diversity time, while political authorization and administrative execution must be calibrated in each institutional scenario; the numerator grows on accumulative development time. If the projection side maintains exponential growth while these constraints persist, the individual cognitive Reynolds number faces long-term upward and divergent pressure. Whether it increases monotonically in every period still depends on the relative movement of the numerator and denominator during that period.

Institutional Level. The pathway of institutional intervention does not run through individual processing capacity—it does not run through each individual's human biological substrate, but through the global effect of an order of prohibition; no separate "institutional cognitive Reynolds number" is posited here, and institutional-level overload pressure must be analyzed according to the complete form of the cognitive Reynolds number in §5.2. Yet the effectiveness of institutional intervention depends on the race between the governance clock and the techno-capital clock.

When the governance clock is shorter than the lock-in period of the techno-capital clock, the intervention arrives before the projection side is locked in—unlocked stock has not formed large-scale sunk costs, narrative load has not sedimented as industry default, and the cost of intervention is low. When the governance clock is longer than the lock-in period of the techno-capital clock, the intervention arrives after the projection side has locked in—sunk costs have already formed, narrative load has already sedimented as industry consensus, and the intervention faces enormous retroactive costs. A ban can stop the increment, but cannot liquidate the stock.

This is the precise formulation of the Collingridge dilemma (Collingridge, 1980) in the MCAF clock language: when intervention is easy, the problem is not yet visible (the debt clock has not yet allowed the damage to manifest); when the problem is visible, intervention has become difficult (the sunk costs of unlocked stock have already locked in).

§3.5 Presentism as a Sociological Symptom of the Temporal Crisis

§3.3–§3.4 argued within the MCAF framework for the structural ordering of the three clocks, the conditions for transition from normalcy to crisis, and the bifurcation of the cognitive Reynolds number at the individual and institutional levels. §3.5 pulls the viewing distance out to a civilizational scale: translating the historico-sociological diagnosis of the rise of "presentism" by Hartog and Li Jing into the MCAF language. Historical sociology has examined temporal regimes linking the present, past, and future independently of this framework. This section uses that discussion to interpret why long-term calibration may give way to immediate readings, rather than treating it as an independent test of the ordering of technological change, governance response, and debt manifestation. Those three periods must still be specified separately in concrete scenarios.

The conditional ordering specified above $\tau_c \ll \tau_g \ll \tau_d$ has a deeper counterpart on the macro-civilizational scale. The historian Hartog (2003/2015) identified that contemporary Western society is undergoing a temporal regime he calls "presentism"—a fundamental change has occurred in the specific, institutionalized relationship among the past, the present, and the future.

Building on this, Li Jing (2023a, 2023b) provided a precise sociological analysis of presentism. She points out that within the progressivist temporal structure of modernity, the present was deprived by the infinitely anticipated future—the present was merely a transition and sacrifice on the way to a better tomorrow. After the collapse of progressivism, presentism simultaneously cancels the importance of both past experience and future expectations; each present is no longer sacrificed in the name of the future, yet the present also erodes and ultimately replaces the past and the future.

Translated into the language of the MCAF framework: presentism is a civilization-level symptom of cognitive debt's transition from concealment to liquidation. When individuals and institutions cease to pursue the long-term calibration of clocks against the sun—abandoning transcendent expectations for the future—and seek only the immediate monetizability of each momentary reading, shadow debt (D_c^{shadow}) continues to accumulate

unnoticed. The gap is no longer tracked over time because the importance of the "sun" has been weakened. The reading of each moment is consumed, traded, and forgotten, replaced by the reading of the next moment. The self-referencing of mapping (already argued in Chapter 1 §4.2—the mapping is rewritten by its use, the clock points to other clocks) unfolds in the temporal dimension as: the reading of this moment points to the reading of the next moment, rather than pointing to any reality that endures across time.

On the economic level, presentism is structurally consonant with the temporal foundation of financialization: extracting and immediately monetizing the value of each present becomes a dominant principle. Li Jing's specific analysis concerns the financialization of the United States after the 1980s and the 'shareholder revolution.' Share price became the ultimate measure of performance; US corporate managers sought to abstract concepts that could be traded on the stock market or used to maintain a favorable market position; and the major industries shaping the US economy were transformed from complex, bureaucratic social entities into dynamic shareholder networks that could be bought, sold, and disassembled. Mapping this US experience onto the temporal enactment of the narrative leverage ratio (Ω) is this book's theoretical extension: within that mapping, the inflation of narrative load (ϕ) stands in tension with the long-term accumulation required by the legitimacy base ($\|L\|$). Its scope must still be tested in each scenario rather than generalized to every enterprise and industry.

On the academic level, Li Jing recounts Hartog's observation that the rise of memory studies and heritage preservation was a reaction against presentism, but was soon absorbed into presentism itself. Memory is a fragment of the past reconstructed around the present identity of the individual, not history based on independent calibration—it is yet another form of mapping's self-referencing, only with the direction shifted from "future" to "past."

This is the temporal-structural foundation of modernity as a civilization-level dimension shift. Accumulative development time drives the exponential inflation of the projection side (Zhao Dingxin, 2019); the positive feedback of triple acceleration makes the inflation self-reinforcing (Rosa, 2013); the rise of presentism marks the completion of the clock dimension's systematic substitution for the time dimension (Hartog, 2003/2015; Li Jing, 2023a, 2023b). These accounts approach the problem from different perspectives, but Hartog and Li Jing stand in an explicit relation of intellectual inheritance. They should therefore be treated as mutually illuminating diagnoses in historical sociology, not as independent empirical confirmations. Together they describe a temporal structure resembling the ordering of the three clocks. The MCAF framework does not take reversal of this tendency as its goal. Its task is to establish a compulsory bookkeeping discipline: under the continuing expansion of the clock dimension, to track the widening gap between that dimension and the dimension of time, and to sound an alarm before the accumulated gap becomes dangerous. This is the operationalized institutional enactment of what von Neumann called "an intelligent exercise of day-to-day judgment."

§3.6 Summary of Variable Time Signatures

Table 1. Time Signatures of the Six Variables

Variable	Temporal Property	Characteristic Time-Scale	Corresponding Clock
Unlocked Stock (C)	Available stock, content quality, and accumulation history jointly constrain narrative load; stock growth is driven by capitalist competition	Accumulation: 10^8 years to centuries; Growth rate: months to years	Techno-Capital Clock
Media Conductivity Coefficient (Λ)	Three-level nested timescales, synergistic amplification	Infrastructure: decades; Distribution Architecture: weeks to months; Contagion Dynamics: hours to days	Techno-Capital Clock
Narrative Load (ϕ)	First-order change is generally easier to observe; second-order change is difficult to audit under uniform coverage and absent contrasts	First order: days to months; Second order: years to decades	Techno-Capital Clock
Meta-Audit Capacity (A)	Slow development, rapid decay; individual level constrained by human biological substrate; the institutional level does not run through each individual's human biological substrate	Development: years to generations; Decay: dependent on task, practice, and support conditions	Governance Clock
Human Biological Substrate (E_c)	Constant	N/A (constant on 10^4 year scale)	Anchor
Legitimacy Base ($\|L\|$)	Slow accumulation, rapid collapse; the first level to reach its scenario-specific failure boundary becomes binding	Accumulation: decades to generations; Collapse: weeks to months	Governance Clock

The governance clock has different meanings at different levels: at the institutional level, it refers to the response cycle of directly modifying the projection side through power; at the individual level, it refers to the micro-level exercise cycle of meta-audit capacity. The core argument of this framework depends on the institutional-level meaning.

Table 2. Temporal Structure of the Three Clocks

Clock	Symbol	Temporal Property	Characteristic Time-Scale	Remarks
Techno-Capital Clock	τ_c	Characteristic period of significant change in the projection-side product; faster growth means a shorter period	Often months; scenario-calibrated	Rhythm of projection-side change
Governance Clock	τ_g	Constrained by evidence, authorization, and implementation procedures; the lower time bound is scenario-calibrated	Often years; scenario-calibrated	Response of Defense Side
Debt Clock	τ_d	Characteristic period from debt sedimentation to first manifestation; at the trigger, the remaining waiting time falls to zero	Scenario-calibrated; decades to generations in some paradigmatic cases in this book, not a universal value	Marker of cognitive-debt manifestation

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§4 The Dynamics of Cognitive Debt

Cognitive debt follows a three-stage lifecycle of "accumulation → concealment → liquidation"—liquidation does not mean the disappearance of debt; it is the forced conversion of the debt form from implicit to explicit.

§4.1 Four Forms

Chapter 1 §4 has already distinguished three layers of irreversibility (§4.1 Information Deficit, §4.2 Dynamic Corrosion, §4.3 Ontological Dimension Reduction). Bookkeeping responds differently to the three layers of predicament—for the first layer, tracking calibration state; for the second layer, tracking corrosion rate; for the third layer, marking the ineliminable fiber. The first three forms are conceptual expressions of the same bookkeeping relation at different temporal resolutions, while shadow debt distinguishes the manifested portion from the not-yet-manifested portion; none of them are formulas into which values can be directly substituted.

Discrete Residual Logic.

$$\Delta D_c = C \cdot \Lambda \cdot \phi - A \cdot E_c \cdot \|L\|$$

This expression is used to diagnose single or non-continuous specific interaction events.

Conceptual Expression of Continuous Flow. When interaction events are sufficiently dense, the discrete residual can be notionally treated as a continuous flow—the system, on the macroscopic scale, continuously bears the velocity difference between the projection side and the defense side. Because the variables are ordinal in nature, this is not written here in infinitesimal calculus form. Its operational meaning is: cognitive debt sediments not only in single events but also accumulates as a continuous flow in high-frequency information environments—the latter corresponds to the phase transition of the cognitive Reynolds number from laminar to turbulent flow

in §5.2. Continuous flow is suitable for the macro-statistical description of information environments, not for single-event diagnosis.

Historical Integral and State Baseline.

$$D_c(T) = D_c(0) + \sum_{i=1}^{N} \Delta D_{c,i}$$

This expression marks the temporal accumulation of cognitive debt. At the meta-framework level, the summation sign records the accumulation of a sequence of events; it does not require cognitive-debt increments (ΔD_c) to be cardinal quantities that can be added across events. Numerical addition becomes meaningful only after a concrete scenario defines a scale. In the Clock Room thought experiment, the expression corresponds to the accumulation of fibers left by historically uncalibrated readings.

Shadow Debt. The difference between the true total amount of cognitive debt (D_c^{true}) and the part currently observable (D_c^{obs}) is defined as shadow debt:

$$D_c^{shadow} = D_c^{true} - D_c^{obs}$$

Observable cognitive debt (D_c^{obs}) is debt that has already manifested in identifiable forms—statistical deterioration of adolescent mental health, book losses on financial products, measurable declines in institutional trust. True cognitive debt (D_c^{true}) is the total amount containing both manifested and unmanifested debt. Shadow debt (D_c^{shadow}) is the difference between the two—the stock that has already sedimented but has not yet triggered visible consequences. It is the most dangerous part of cognitive debt: not because it is larger, but because it is invisible. Unbooked debt triggers no response from the defense side within the system—this is precisely why the framework requires the active registration of shadow debt.

The existence of shadow debt makes the debt clock (τ_d) argued in §3.3 the most unpredictable of the three clocks—the analyst knows that shadow debt (D_c^{shadow}) is accumulating there, but cannot predict when and in what manner it will manifest as observable cognitive debt (D_c^{obs}). In the financial domain, the trigger of manifestation can be a default; in the AI domain, it can be a large-scale hallucination propagation; in the physical environment, it can be the irreversible compression of an aquifer. Before the trigger moment, shadow debt is invisible to participants within the system—this is the cognitive-debt version of the Collingridge dilemma argued in §3.4.

The increment of cognitive debt can be negative—when the defense side operates effectively (institutional calibration recovered, meta-audit capacity rebuilt), previously sedimented Layer-1 debt (uncalibrated readings) and Layer-2 debt (corroded mappings) can be partially repaid. But Layer-3 debt (dimension loss) is irreparable—dimensions discarded by the mapping are not restored by subsequent auditing. The "liquidation" in §4.2 below specifically refers to the forced conversion of implicit debt into explicit loss; it is a morphological conversion, not a stock write-off.

§4.2 The Three-Stage Lifecycle

Accumulation Stage. Each unit of positive cognitive debt increment sediments within the system through the historical integral; the high narrative load of the projecting party obscures the visibility of the liability—the

techno-capital clock is running while the governance clock has not yet responded.

Concealment Stage. Cross-period feedback mechanisms give accumulated debt a "compounding" character: cognitive debt not liquidated in the current period continually compresses the processing capacity available for audit, reducing the effective exercise of meta-audit capacity (A) and accelerating new debt. At the same time, the projection side supplies low-friction substitutes in the outer layers—the automation paradox renders operational friction obsolete, while algorithmic *xiangyuan* displaces affective friction—inducing subjects to shift investment from the core zone outward and causing meta-audit capacity to atrophy endogenously as its field of exercise disappears. The human biological substrate (E_c) remains approximately constant; what deteriorates is the processing capacity and field of work available to meta-audit, and these changes are recorded in the effective exercise of meta-audit capacity rather than counted again under both A and E_c. During this period, cognitive debt exists as implicit friction—fragmented attention, degraded judgment, and eroded institutional trust—but has not yet produced systemic explicit losses: debt has sedimented, but the debt clock has not yet come due.

The Covering Law: Daoist Time as an Endogenous Pressure against Perpetual Concealment. The concealment stage of cognitive debt cannot continue indefinitely without qualification because it is subject to a covering law. Zhao Dingxin (2019) calls this "Daoist time" (an idea whose intellectual lineage can be traced back to the insight of Laozi's "Reversion is the action of the Dao" (*Tao Te Ching*, Chapter 40; Lynn, 1999)—Zhao developed it into an explicit sociological proposition) and gives it a precise sociological formulation: for social organizations, ideas, and institutions of any nature, as they become powerful, the social forces and social mechanisms that weaken them will also become increasingly important.

Zhao Dingxin distinguished the fundamental difference between this law and the mechanical cyclical theories common in Western social science (such as Olson's interest group theory and Turchin's cohesion cycles): the latter hold that the causes behind history moving "up" and moving "down" are the same, whereas Daoist time holds that the causes of upward and downward fluctuations are sometimes the same and sometimes different. "The Dao that can be described in language is not the constant Dao." The speakable Dao(s), that is, specific social mechanisms, all change with historical context and are not the "constant Dao." The "constant Dao" is the covering law itself: success endogenously manufactures the conditions of its own weakening.

This covering law has a precise counterpart in the three-level structure of the MCAF framework. The exponential inflation of the projection side ($C \cdot \Lambda \cdot \phi$) continuously accumulates cognitive debt during the concealment stage. But as the projection side becomes powerful, three categories of weakening pressure can be distinguished, with the third requiring additional conditions of transformation: First, the success of the projection side increases its exposure surface to being imitated and gamed—once the first derivative of narrative load (ϕ) becomes visible, it triggers adaptive drift (already argued in Chapter 1 §4.2). Second, the expansion of the projection side consumes the buffer of the legitimacy base ($\|L\|$)—the short-term dividends of performance legitimacy mask the long-term erosion of procedural legitimacy, but the masking itself consumes the capacity to mask. Third, the projection side's deepest effect on the defense side—filling the bandwidth with noise[1] (§6 Postulate 2)—

[1]This book uses "bandwidth" only in the information-theoretically measurable sense: the amount of information processable per unit time. This delimitation is not of this book's own making—the cited literature itself separates it from the rhetorical usage: Zheng and Meister compute the information capacity of neurons by the Shannon–Hartley theorem while devoting a dedicated section to refuting "not enough bandwidth"-style claims. The original text of that section: 6.3 The Musk illusion. Many people believe that their internal life is much richer than anything they can express in real time through their mouth or otherwise. One can view this illusion as a version of subjective inflation (see 5.2) generalized beyond visual perception: Because we could engage in any one of the 2^{10} possible actions or thoughts in the next second, it feels as though we could execute them all at the same time. In practice, however, they happen

reaches saturation upon success: bandwidth is a finite physical constant, and the marginal effect of noise no longer increases once the bandwidth is saturated. Yet saturation by itself constitutes no counterpressure against concealment—the occupied processing capacity does not release itself merely because saturation is reached, and meta-audit capacity still lacks room to operate while that capacity remains occupied (§5.3); a suppressed need for audit does not automatically become auditing either. It must be converted into actual work through independent evidence, organizational support, or new conditions for audit work, and only then does it exert counterpressure against the concealment mechanism. Whether these conditions of transformation are met must be specified in the concrete scenario; it does not follow from the diminishing marginal effect of noise alone.

Daoist time therefore constitutes one structural pressure moving cognitive debt from concealment toward liquidation. It is not a specific trigger event, but a consequence of the covering law that success gradually produces conditions for its own weakening. The "baroque" degeneration observed by von Neumann in the field of mathematics (already argued in Chapter 1 §2.2) is a manifestation of the same law at the disciplinary level: a discipline becomes ever more refined but ever more degenerate after straying far from its empirical source, until the only remedy is a return to the source—this is liquidation.

Liquidation Stage. When the accumulated debt breaches the stress limit of the legitimacy base, or an exogenous shock forcibly realizes implicit liabilities—the debt clock comes due; implicit debt is forcibly manifested—cognitive debt converts from book-entry implicit debt into explicit damage. The 2008 subprime crisis is a typical case of the liquidation of cognitive debt in the financial system: rating agencies lost their auditing independence due to structural conflicts of interest (the effective capacity of the meso-level meta-audit link declined sharply); complex structured products used layer-upon-layer packaging to present high-risk assets through a highly rated narrative (narrative load far exceeded the legitimacy base); when underlying assets deteriorated synchronously, the crisis erupted in full force—the International Monetary Fund estimated in April 2009 that cumulative potential write-downs on mature-market credit by global market participants over 2007–2010 would amount to approximately USD 4.1 trillion, of which banks would bear roughly USD 2.5 trillion. Including an additional estimated USD 340 billion in write-downs on emerging-market asset exposures brings the global banking total to approximately USD 2.8 trillion. These were estimates at the time, not finally realized losses (IMF, 2009).

The Manifestation Function $M(\tau)$. To mark the mechanism by which shadow debt D_c^{shadow} is converted into observable debt D_c^{obs} at a trigger moment τ^*, this scenario-specific transformation is denoted by the **manifestation function** $M(\tau)$. Consistent with the identity of the meta-formula stated in §2.0, $M(\tau)$ is a structural marker for classificatory diagnosis, not a solvable analytic dynamical system; its domain, range, and specific form must be supplied by the scenario model. In finance, $M(\tau^*)$ may unfold as the institutionalized endogenous chain "underlying asset default—margin call—ratings downgrade." In physical environments, it may unfold as irreversible aquifer compression or an ecological threshold transition. What the AI domain lacks is this financial-style institutionalized, endogenous, and compulsory manifestation chain—not observable harms such as large-scale hallucination propagation or erroneous decisions (see Chapter 15, §3). These domains share the

sequentially. For the most part, this is a harmless illusion. However, when paired with the immense fortune of Elon Musk, the belief can lead to real-world consequences. Musk decided to do something about the problem and create a direct interface between his brain and a computer to communicate at his unfettered rate: "From a long-term existential standpoint, that's, like, the purpose of Neuralink, to create a high-bandwidth interface to the brain such that we can be symbiotic with AI," he said. "Because we have a bandwidth problem. You just can't communicate through your fingers. It's just too slow." Based on the research reviewed here regarding the rate of human cognition, we predict that Musk's brain will communicate with the computer at about 10 bits/s. Instead of the bundle of Neuralink electrodes, Musk could just use a telephone, whose data rate has been designed to match human language, which in turn is matched to the speed of perception and cognition. (Zheng & Meister, 2025) This book takes the former layer, not the latter.

abstract structure in which shadow debt becomes observable after a trigger, but the framework does not presume that every manifestation must take the form of a discontinuous jump. $M(\tau)$ does not predict τ^*; it only marks the transformation mechanism that actually operates in a specific scenario.

It must be strictly declared: the liquidation of cognitive debt does not equal the disappearance of debt—it is the forced conversion of the liability form (from implicit to explicit), not the write-off of the book entry. What the system enters after liquidation is not a "zero-liability" state, but a state in which the structural friction previously obscured by narrative has been physically exposed at the cost of explicit loss.

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§5 Three Derived Indices

To perform cross-period diagnosis on the "accumulation—concealment—liquidation" dynamics described above, it is necessary to derive operable analytical tools from the meta-formula. The following three derived indices provide structural comparison benchmarks for the empirical mapping in the subsequent chapters, from the three orthogonal dimensions of name-reality deviation, flow regime determination, and defense-line conservation, respectively.

§5.1 Narrative Leverage Ratio (Ω)

The narrative leverage ratio $\Omega = \phi/\|L\|$ measures the structural pressure of narrative relative to the legitimacy boundary. Leverage is not itself pathological; after a specific scenario defines the relevant scale, a rupture in the trust chain may trigger forced liquidation if narrative load continues to rise while the legitimacy base stagnates.

Borrowing the structural intuition of financial leverage (total assets / equity ratio): the legitimacy base is equivalent to the system's real "equity"—the boundary of substantive or institutional support that narrative can obtain; narrative load is equivalent to the "total assets" in current operation of the system—the total amount of narrative actually released by the projecting party. The part exceeding the legitimacy boundary constitutes "narrative debt" lacking substantive support:

$$\phi - \|L\| = \|L\| \cdot (\Omega - 1)$$

The dynamic states of the narrative leverage ratio are divided into three. **Name-Reality Correspondence ($\Omega \approx 1$):** Narrative load matches the material or institutional value it provides, fully supported by the legitimacy base. **High Leverage and Smooth Reversion:** The system enters a high-leverage state during a specific period, but if the excess narrative ultimately transforms into actual delivered value increments (the legitimacy base substantively thickens), the narrative leverage ratio will return to a steady state. **Excessive Leverage and Forced Liquidation:** If narrative load continues to expand in one direction while the legitimacy base stagnates or undergoes endogenous erosion, a shock that breaks the trust chain may trigger forced liquidation of cognitive debt.

In the language of the Clock Room thought experiment, the narrative leverage ratio compares the relative pressure of readings against the conditions of calibration. Only after a scenario defines and normalizes its scale can 1 serve as a reference for correspondence between name and reality. At the meta-framework level, $\Omega \approx 1$ and $\Omega \gg 1$ denote relative states rather than universal cross-scenario thresholds.

§5.2 The Cognitive Reynolds Number (Re_c)

Borrowing the structural intuition of the Reynolds number in fluid mechanics (the ratio of inertial force to viscous force), the cognitive Reynolds number (the full term is "quasi-cognitive Reynolds number," marking the distinction from the physical Reynolds number; the running text abbreviates it as the cognitive Reynolds number) Re_c is an order-of-magnitude reference comparing the projection side's information-arrival pressure with the defense side's processing capacity; its complete form is the ratio of the projection side's total load to the defense side's total processing capacity.

$$Re_c = \frac{C \cdot \Lambda \cdot \phi}{A \cdot E_c \cdot \|L\|}$$

At the micro-individual level, unlocked stock, narrative load, the legitimacy base, and meta-audit capacity are provisionally treated as background conditions within a specific short-term slice and absorbed into the scenario scale, yielding the following heuristic simplification. This simplification applies only to within-scenario comparison over a period of months to years; cross-period or macro analysis must return to the complete form:

$$Re_c \approx \frac{\Lambda}{E_c}$$

The numerator is the media conductivity coefficient, which can be operationalized as the density of information arrival per unit time under a specific media environment. The denominator is the human biological substrate, corresponding to the order-of-magnitude reference for conscious human information processing—approximately 10 bits/s (Zheng & Meister, 2025), with a working memory of approximately 4 ± 1 chunks (Cowan, 2001). The human biological substrate is approximately constant on the time-scale of civilizational history. When the cognitive Reynolds number lies within a lower interval calibrated for the scenario, information flow remains within the host's processing threshold and the system is in a "laminar" state capable of sustaining meta-audit. When it crosses that scenario-specific critical interval, information-arrival pressure exceeds available audit capacity, the system enters "turbulence," the host finds item-by-item auditing difficult to sustain, and unprocessed informational redundancy may sediment as cognitive debt.

In the language of the Clock Room thought experiment, the cognitive Reynolds number measures the ratio between the arrival density of clock readings and the subject's capacity to calibrate readings. Once a concrete scenario defines and normalizes a ratio of information arrival to auditable capacity, 1 may be designated as that scenario's capacity boundary. A value below the boundary indicates that the subject can still calibrate readings individually; a value above it indicates that readings arrive faster than the available calibration capacity, making reliance on Strategies 1 through 4 from Chapter 1, §1.3—taking a statistic, looking for clustering, observing running rate, and selecting an authoritative clock—more likely. Here, 1 is a scenario-specific normalization, not a universal cross-scenario constant.

The Triple Structure of Acceleration. Along the modern media trajectory examined in this chapter, the numerator of the cognitive Reynolds number (Λ) is subject to persistent upward pressure, and this change is not a simple linear process. Rosa (2013) identified three mutually reinforcing dimensions of acceleration: technological acceleration increases the physical arrival density of information (corresponding to the macro-infrastructure layer of the media conductivity coefficient Λ), the acceleration of social change shortens the replacement cycle

of institutions and relationships (corresponding to the meso-distribution architecture layer of the media conductivity coefficient (Λ)), and the acceleration of the pace of life forces individuals to process more information in less time (directly compressing the time available for auditing under the constraint of E_c). The three constitute a positive feedback loop—acceleration in any single dimension reinforces the acceleration pressure on the other two dimensions.

This means that the cognitive Reynolds number faces strong nonlinear upward pressure: under the combined effects of triple acceleration, the media conductivity coefficient (Λ) may grow at a superlinear rate, while the human biological substrate (E_c) remains approximately constant on the time-scale of civilizational history. While these conditions persist, the long-term tendency is upward; whether the trajectory is monotonic in every period still depends on the relative movement of the factors within the scenario.

Rosa's analysis also reveals a counter-intuitive phenomenon that the MCAF framework needs to incorporate: technological acceleration should increase the individual's free time (faster tools should make people more idle), but the actual pace of life has instead quickened. Translated into the language of the MCAF framework: the efficiency gain on the projection side did not release the defense side's bandwidth—it manufactured new demands to fill the released bandwidth. This is the confirmation, from the perspective of social acceleration, of §5.3 (*Tao Te Ching*, Chapter 11: "Therefore this is how what is there provides benefit and how what is not there provides functionality"; Lynn, 1999): the capacity of the projection side to eliminate "nothingness" (*wu*) is accelerating, and the "nothingness" (unoccupied bandwidth) on which the defense side depends is continuously shrinking.

It must be declared: the cognitive Reynolds number is a heuristic ordinal comparison tool, not a precisely calculable physical dimensionless number. Its core value lies in establishing a structural baseline for comparison across eras and media—the human biological substrate is approximately constant, while the media conductivity coefficient faces an overall upward pressure along the media trajectory examined here—rather than in providing an absolute numerical value.

§5.3 Cognitive Friction (μ_c)

Cognitive friction $\mu_c = A \cdot E_c$ is not a design flaw—it is a necessary condition for maintaining agency, and its absence has two independent paths: the automation paradox renders operational friction obsolete, and algorithmic *xiangyuan* displaces affective friction.

After reintroducing meta-audit capacity as a dynamic variable into the simplified form, the denominator of the cognitive Reynolds number expands into cognitive friction:

$$\mu_c = A \cdot E_c, \qquad Re_c \approx \frac{\Lambda}{\mu_c}$$

In fluid mechanics, dynamic viscosity characterizes the internal friction of a fluid—the greater the viscosity, the smaller the Reynolds number. Cognitive friction occupies precisely this dynamic denominator position.

Cooper (1999) used "cognitive friction" to describe the resistance encountered when people engage with complex systems whose rules change with the problem, while explicitly allowing that small amounts need not be harmful. MCAF uses the term for the effort required for independent verification, reflection, and relational work, treating this audit friction as necessary for agency and cognitive health; this is not a simple reversal of

every value Cooper assigned to friction. Chapter 2 §3.1 provides a philosophical reference for understanding this necessary effort: Laozi's "Therefore this is how what is there provides benefit and how what is not there provides functionality" (*Tao Te Ching*, Chapter 11; Lynn, 1999)—what the projection side provides is "having" (content, information, services); what the defense side depends on is "nothingness" (unoccupied bandwidth, unsubstituted friction, unenclosed channels). Eliminating cognitive friction is equivalent to filling the empty cavity of the bowl with clay—the bowl remains, but it can no longer hold anything.

Meta-audit capacity and the human biological substrate must be simultaneously present to generate effective friction. The human biological substrate itself does not fall to zero; when processing capacity available for audit is fully occupied by non-auditing tasks, meta-audit capacity has no room to operate. When meta-audit capacity is zero—no work is performed—the biological substrate supplies only processing capacity and produces no audit, so readings pass straight through.

The absence of cognitive friction has two independent paths. **The first: the automation paradox renders operational friction obsolete.** Bainbridge (1983) named this the "ironies of automation": the more advanced the control system, the more crucial the human operator's contribution—yet the operator's skills degrade from lack of use. This chapter calls its extension to the meta-audit level the automation paradox: the more automated and reliable the system's operation, the more the opportunities and capacity of the human participant to sustain meta-audit atrophy. Automation encapsulates complex processes, which originally required the subject to engage instrumental rationality in gaming them, into "one-click" transparent delivery; the long-term absence of operational friction leads to degenerative deskilling, and cognitive friction atrophies first in the instrumental dimension. **The second: algorithmic *xiangyuan* displaces affective friction.** The algorithm exploits the property that non-human objects are far more compliant than real persons, masquerading as a forever-agreeing digital agent—Confucius's verdict on the *xiangyuan*: "Your good, careful people of the villages are the thieves of virtue" (*Analects*, "Yang Huo"; Legge, 1893)[2]—and displaces the affective friction on which affect audit depends for calibration with a zero-friction digital substitute. Thus, the subject and the projection side form a structure analogous to a "compliant equilibrium": in the short term, accepting a compliant narrative reduces the subject's cost of judgment, while the projection side obtains higher transmission efficiency with lower resistance. The game-theoretic vocabulary here describes a mutually reinforcing relation; it does not claim that a complete strategy set, payoff ordering, or equilibrium proof has been supplied. The micro-mechanisms of the two paths—how the automation paradox advances from the outer layer to the core zone, and how algorithmic *xiangyuan* systematically expands the population size "needing a digital artificial kidney"—will be unfolded in Chapter 5 using Hsu's PSH theory.

Experiments on complex-estimate audits also provide countervailing evidence. In an experiment with 94 senior-level auditors from three Big 4 firms, Griffith, Hammersley, Kadous, and Young (2015) show that a deliberative-mindset intervention improves auditors' ability to identify unreasonable estimates by changing how they process information rather than merely increasing their work effort. Bucaro (2019) further shows that simply emphasizing the use of professional judgment is insufficient to reduce mechanistic decision-making, whereas a systems-thinking perspective improves auditors' evaluation of complex situations and their application of professional judgment. These findings do not establish that similar interventions are equally effective in every cognitive task, but they show that meta-audit capacity can be improved through training, process redesign, and cross-source

[2]The original text of the *Analects*, "Yang Huo," writes the term as 乡原, where 原 is interchangeable with 愿; the *Mencius*, "Jin Xin II," quotes the same form. This book's conceptual terminology uniformly uses "algorithmic *xiangyuan*" (算法乡愿), while quotations follow the original-character form.

information integration rather than being confined to decay along the paths of automation and compliance.

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§6 Five Structural Postulates

The five postulates are not arguments—they are statements of structural premises required for the framework to operate. The previous chapter supplies the epistemological basis concerning lossy mapping and the limits of calibration; the social-mechanism premises used here are supported separately by the literature and chapters cited below.

The following five postulates are the structural premises on which the operation of the meta-formula depends. They do not assert necessity—social systems do not obey the determinacy of natural laws—but argue structural tendencies: the direction toward which the system is most likely to slide in the absence of conscious intervention. The postulates share the epistemological basis supplied by the previous chapter, but their social-mechanism premises were not independently established there. This section states each postulate, identifies MCAF's specific increment, and points to the corresponding literature and chapter-level arguments.

§6.1 Postulate 1: Technological Expansion Is the Product of Competitive Power-Selection

Technological expansion is not natural evolution, but follows the "artificial variation–selection–inheritance" mechanism of power-selection within Tang Shiping's social evolution paradigm (Tang, 2013). Hoffman's tournament model identifies one dynamic of competitive power-selection: when spoils are highly skewed, the parties' capabilities and mobilization conditions are relatively close, and the gains from peace are difficult to divide, rulers are more likely to sustain investment in war and military technology; under other cost and capability conditions, peaceful equilibria remain possible (Hoffman, 2015). Zhao Dingxin confirmed the contingency of this mechanism from the perspective of institutional competition: industrial capitalism is not an endogenous necessity of any civilization, but a contingent product of Europe's multi-state competition system (Zhao Dingxin, 2019). The eighty-odd-year selection history of AI routes from 1943 to 2026 is the concrete unfolding of this postulate in the technological domain—four selection mechanisms (national security and bureaucracy, academic factions and paradigms, capital channels and interest rates, and the orthogonal selection of substitute arenas), together with their superposition, substitution, and orthogonal relations, are treated in detail in Chapter 34.

MCAF Increment. Packaging this contingent product of power into a narrative of "universal human progress" is itself an academic institutionalization with a high narrative leverage ratio. The fundamental asymmetry between the accumulation time-scale of the four categories of ordered structures (material, informational, biological, attentional) (hundreds of millions of years to centuries) and the unsealing time-scale (years to months) provides the potential energy basis for the exponential inflation of the projection side (§3 Variable Time Signatures).

§6.2 Postulate 2: The Incommensurability of Local Optimization and Global Coordination

The human biological substrate is a product of long evolution. This framework further postulates that the coupling of perception, action, and regulation cannot be adequately represented by any single local task metric. Von Neumann pointed out that the brain's system of notations is "not digital but statistical" (von Neumann, 1958). When a local optimization algorithm is forcibly mapped onto this globally coordinated bio-cognitive

substrate, the two may lack a common unit of measure. The structural residual produced by this mapping is this book's ontological postulate concerning cognitive debt, not a conclusion that follows from notational differences alone.

Chapter 1, §4.3 (Ontological Dimension Reduction) shows that, under the usual conditions of open Euclidean regions or finite-dimensional manifolds, a continuous mapping from a higher-dimensional true state space to a lower-dimensional observation space cannot remain injective across all true states. At least one non-singleton fiber therefore exists. This limits recovery of the complete state from that mapping; if a task requires only a defined low-dimensional quantity, whether the mapping preserves the information needed for that task must be assessed separately.

MCAF Increment. The order-of-magnitude reference of the human biological substrate (approximately 10 bits/s) will not be changed by the technological environment—it is a biological constraint determined by the current best empirical estimate. But the replacement by the technological environment manufactures a massive amount of noise, causing the effective signal-to-noise ratio of those 10 bits/s that is available for meta-audit to drop sharply: variable-ratio reinforcement reshapes the dopamine system so that attention is occupied by stimuli unrelated to auditing, and fragmented information erodes the duration of deep attention. What degrades is not the physical bandwidth of the human biological substrate, but the effective working time of meta-audit capacity on that bandwidth. The deepest pathway of the projection side's effect on the defense side is not to break the defense line from outside, nor to compress the bandwidth itself, but to fill the bandwidth with noise—keeping the axle hole of the cognitive wheel present yet clogged with foreign matter. The step from differences in notation to structural costs between local optimization and overall coordination is a postulate of this framework. It does not follow from notation differences alone; the omitted coordination requirements and their consequences must be specified in each scenario.

§6.3 Postulate 3: Meta-Audit Is Continuous Work against Entropy Increase

The default state of the cognitive system is to rely on the cemented legitimacy base for automated pattern recognition—the coordinate system collapses to zero. Meta-audit capacity demands breaking the default, establishing an orthogonal coordinate system independent of the narrative closed-loop of the projecting party. This is a high-dissipation behavior contrary to the instinct for cognitive minimalism—once the work stops, the system slides back into blind automation. Von Neumann gave the engineering formulation: the only possible safety mechanism depends on "an intelligent exercise of day-to-day judgment" (von Neumann, 1955).

The role of meta-audit in the three layers of irreversibility differs: in the first layer (Information Deficit), introducing an external reference to shrink the fiber; in the second layer (Dynamic Corrosion), repeatedly recalibrating to counter the degradation of the mapping; in the third layer (Ontological Dimension Reduction), admitting that the mapping is lossy and marking the ineliminable residual.

MCAF Increment. When the projection side enters a range of exponential growth, the subject does not simply stop working; rather, the projection side systematically removes the conditions required for that work—algorithms use extremely rapid concurrent distribution to manufacture narrative capture (media conductivity coefficient inflation), platforms use zero-friction delivery to deepen the automation paradox (role audit degrades), AI uses compliant output to displace affective friction (affect audit is abandoned). The decay path of meta-audit capacity is smooth, gradual, self-concealing—in contrast to the threshold fracture of the legiti-

macy base (Postulate 4).

§6.4 Postulate 4: The Legitimacy Base Accumulates Slowly but Can Collapse Rapidly, and the Succession of Diversity Is Constrained by Scenario Conditions

The construction of the legitimacy base depends on long cross-generational games; its collapse manifests as a sharp rupture after a threshold is breached. Tang Shiping pointed out: "while institutions often are relatively stable, they are subject to change" (Tang, 2011). This temporal asymmetry means that the legitimacy base is a rigid constraint in the short term and cannot, like narrative load, inflate exponentially at near-zero marginal cost.

The legitimacy base faces pressure from two directions. **Internal:** The more a legitimacy dimension is explicitly quantified and anchored as a single decision target, the more exposed it may become to adaptive drift and indicator corrosion. Chapter 1, §4.2 develops the relevant mechanisms through five research traditions. Goodhart (1984) and Campbell (1979) provide mechanism-level references for the risk of distortion as control pressure and indicator use increase, while Zhao Dingxin (2022) supplies a methodological reference for the interpretive gap introduced by discretization. These sources do not make distortion an unavoidable necessity. Multiple indicators explicitly recognized as imperfect, external oversight, and institutional design may mitigate the risk, whose actual magnitude must be tested in each scenario. **External:** The diversity of narrative load can be mass-produced at near-zero marginal cost. The parts of the legitimacy base that concern value identification and ideological diversity are constrained by diversity time; changes in political and administrative institutions require scenario-specific analysis. Under Ashby's (1956) Law of Requisite Variety, this gap makes complete regulation by the defense side difficult.

MCAF Increment. In a specific scenario, bodily autonomy may be mapped to the physical boundary at the micro level and is not governed by the anchoring paradox—a wall cannot be "gamed." Meso-level institutional rules and macro-level consensus structures, by contrast, are exposed to adaptive drift. This scenario mapping does not introduce a third hierarchy: the classification still follows the separate level and category axes defined in §2.6, with the first level to reach its scenario-specific failure boundary becoming binding. Unlike the smooth decay of meta-audit capacity, failure of the legitimacy base takes the form of threshold fracture.

§6.5 Postulate 5: The Topological Structure of Historical Scenes Allows Cross-Period Pattern Recognition

Although technological forms and narrative content undergo drastic changes in history, the qualitative phase trajectories of the projection-defense dynamic can belong to the same structural type across different historical periods. "Cross-period isomorphism" lies not in the numerical convergence of specific parameters, but in whether the causal driving variables, coupling structures, and temporal scale asymmetries are topologically equivalent. The "model arbitrage" of financial markets and the "syntactic arbitrage" of complex technological systems share the same fast variable (the exponential inflation of narrative load) and the same slow variable (the slow evolution of the legitimacy base), employing the high-narrative-leverage logic of "extremely complex input, extremely certain output," realizing the privatization of gains and the socialization of risk by masking uncertainty.

MCAF Increment. The dynamic structure described in Postulate 4 is jointly constrained by two parameters operating on different time-scales: narrative load grows on accumulative development time, while the legitimacy

base faces internal erosion from the anchoring paradox and external constraint from diversity time. This chapter uses the fast-slow relation as a heuristic analogy to the cusp catastrophe in Thom's catastrophe theory (Thom, 1975); Zeeman applied cusp-catastrophe classification to stock-market crashes (Zeeman, 1974).[3]

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§7 Epistemological Delimitation and Operationalization Boundary

At the meta-framework level, the variables express relative scale, direction of change, and structural pressure. The meta-formula is a bookkeeping template rather than a predictive tool; projection mapping is a falsifiable empirical hypothesis, and its baseline must remain open to dynamic re-audit.

§7.1 The Mathematical Meaning of the Variables and the Residual

The meta-formula is not an "explanation by law" in the sense of classical physics, but an ideal-typical space for structural-mechanism narratives in the social sciences. At the meta-framework level, multiplication, subtraction, summation, and ratios are structural encodings analogous to dimensional analysis. They express relations and directions among variables but do not supply cardinal scales directly comparable across scenarios. These symbols acquire numerical meaning only after a scenario specifies variable types, scales, and aggregation rules.

The minus sign marks the direction of the residual produced by the collision of two heterogeneous structures. High-frequency discrete information flows cannot be processed losslessly by the human continuous-statistical cognitive substrate, so their intersection produces a mapping residual. At the meta-framework level, cognitive debt denotes the structural cost of this incommensurability, not the arithmetic difference between two naturally commensurable quantities.

The Scenario-Dependence of Variable Types. §2.0 has already declared that the meta-formula is an interdisciplinary chart of accounts and that variable types are not specified at the meta-formula level. The same variable may be mapped to different mathematical types in different scenarios: the media conductivity coefficient (Λ) may be operationalized as information-arrival density per unit time in a micro-level diagnosis of the cognitive Reynolds number, whereas in the three-level analysis of narrative arbitrage it may unfold into macro, meso, and micro components; meta-audit capacity (A) may be represented by a scalar proxy at the individual level and by a three-level structure combined with role/affect categories at the institutional level. The concrete mathematical type and aggregation operation must be specified by the scenario model and cannot be guaranteed in advance by the meta-formula.

[3]The application of catastrophe theory in the social sciences has been criticized for its oversimplification of complex systems (see Sussmann & Zahler, 1978). The MCAF framework takes a cautious position: it does not claim that the liquidation of cognitive debt strictly follows the mathematical form of the cusp catastrophe, but only borrows its topological intuition that "the interaction of the normal factor and the splitting factor leads to threshold fracture," as a **heuristic model** for the dynamic structure of Postulate 4. The exponential inflation of narrative load corresponds to the normal factor of the cusp catastrophe (driving the system to slide smoothly along the equilibrium surface), and the slow evolution of the legitimacy base corresponds to the splitting factor (determining when the system falls off the edge of the cliff)—under the scenario-classification and path conditions specified below, a small change in the control parameters that crosses the fold boundary terminating the occupied stable branch triggers the corresponding jump. The jump itself is what the cusp structure entails mathematically; whether the jump constitutes an irreversible loss is a Layer-3 judgment of this book (dimension loss is irreparable), to be assessed separately against the scenario's conditions for state recovery. If the dynamics of a particular scenario are independently tested and shown to conform to the control-parameter structure of a cusp catastrophe, and the path of its control parameters crosses the fold boundary at which the currently occupied stable branch terminates, then "threshold fracture"—the sudden forced liquidation of cognitive debt after slow accumulation—is, within that scenario, a mathematical entailment of the structure rather than an accidental observation requiring independent proof case-by-case. Both the scenario classification and the path identification are falsifiable empirical hypotheses (§7.2 Level 2), and the entailment holds only when both are valid.

The Operational Boundary of the Meta-Formula. The meta-formula specifies three structural relations in advance: product-form aggregation on each side, the direction of the residual between the two sides, and the minimum constraint on the legitimacy base. It cannot be directly differentiated, integrated, or solved for equilibrium; such operations belong to scenario models with defined scales. The meta-formula is therefore more structured than pure narrative but cannot generate predictive trajectories. Its epistemological status resembles dimensional analysis: it constrains which relations must be recorded without claiming strict mathematical consequences in the absence of scenario-specific scales.

§7.2 The Three-Level Epistemological Demarcation

This framework rejects the non-falsifiability of grand theory and establishes a three-level epistemological demarcation to ensure auditability.

Level 1: The Meta-Formula Is an A Priori Bookkeeping Discipline. "The difference must be recorded" is not itself an empirical proposition and therefore is not directly falsified by an empirical counterexample. It does not provide specific empirical propositions, nor does it predict when a crisis will occur; rather, it establishes a questioning discipline: the analyst cannot discuss only techno-capital efficiency (the projection side) without asking whether the meta-audit mechanism (the defense side) is absent; they must assess whether the techno-capital period is shortening and how large the gap is between the techno-capital and governance periods. This questioning discipline is uniformly applicable across disciplines—regardless of whether the analytical object is an AI system, a financial product, or a car's hidden door handle, the same set of structural questions (Is the projection-side product inflating? Which factor on the defense side is weakest? What is the ordering of the three clocks?) must be compulsorily posed.

Level 2: Projection Mapping Is a Falsifiable Empirical Hypothesis. Mapping the things in a concrete real-world scenario into the variable drawers of the meta-formula is a synthetic empirical act. "The difference must be recorded" is non-falsifiable, but "how to define real things and place them into the corresponding variables" is entirely falsifiable. If the mapping conflicts with reality, what is wrong is the category classification of the bookkeeping, not the bookkeeping law.

Level 3: The Mapping Must Accept Dynamic Re-Auditing. With the iteration of projection-side strategies, the original mapping may at any time undergo category misplacement or dimension omission; the framework requires the user to perform continuous re-auditing of their mapping baseline.

§7.3 Three Kinds of Irreversibility Facing the Framework Itself

The previous chapter's judgment that the clock is not time also applies to the MCAF framework itself. The MCAF framework is a mapping function; it maps cognitive reality into readings composed of six variables. It therefore faces three kinds of irreversibility.

First Kind. The empirical calibration of MCAF's six variables depends on external data—compute tracking, transaction data, and crash-test reports. These data are themselves outputs of other mappings, so their sources, errors, and independence from the audited system must be examined. Only when the data are generated within the same narrative closed loop and fail to impose independent constraints does MCAF degenerate into "calibrating a clock with a clock."

Second Kind. If MCAF is used as a policy tool and the cognitive Reynolds number or narrative leverage ratio

enters rewards, sanctions, or resource allocation, regulated parties face incentives to adjust their behavior around those indicators. Readings may then improve without a corresponding improvement in underlying cognitive health; this means that MCAF's mapping function will be rewritten through its use, and MCAF's mapping baseline must therefore remain subject to continual re-audit.

Third Kind. The MCAF framework is a product of language. Six variables, the product structure, the clock ordering—these are all operations that project high-dimensional cognitive reality onto a low-dimensional framework processable by the human biological substrate. Dimension reduction is unavoidable. The framework captures certain dimensions (the power comparison between projection and defense, the mismatch of temporal scales), but necessarily loses other dimensions (the micro-dynamics of affect, the texture of cultural difference, the incommensurability of individual experience).

The framework is soberly self-aware of this. This is precisely the meaning of this section: the MCAF framework is not exempt from its own diagnosis. It is a clock—possibly a useful clock, because it compels the user to perform bookkeeping. But it does not represent real experience. The user of the MCAF framework should annotate beside every application: This is not cognitive reality; this is a bookkeeping framework about cognitive reality. The value of the framework lies not in its equivalence to reality, but in the fact that it marks the existence of the fiber more honestly than when there is no framework at all.

§7.4 Transition of Proxy Indicators

Starting from this chapter's ontological definitions of the six variables, the following table lists proxy indicators that may be used in empirical measurement and provides a reference for scenario mapping in later chapters.

Table 3. Proxy-Indicator Directions for the Six Variables

Variable	Definition Location	Proxy Indicator Direction (Examples)	Measurement Nature
Unlocked Stock (C)	§2.1	Fossil fuel extraction volume, computing power capital expenditure (an input proxy, to be distinguished from stock), training data corpus scale, genome sequencing throughput	Ordinal • Cross-Period Trend
Media Conductivity Coefficient (Λ)	§2.2	Information arrival density per unit time, Platform Daily/Monthly Active Users (DAU/MAU), Algorithmic recommendation personalization precision	Ordinal • Cross-Period Trend
Narrative Load (ϕ)	§2.3	Scope of claims and commitments, degree of disclosure of key limitations, difficulty of independent verification, Advertising expenditure as % of revenue, Financial product nesting layers, AI model opacity	Ordinal • Structural Comparison

Variable	Definition Location	Proxy Indicator Direction (Examples)	Measurement Nature
Meta-Audit Capacity (A)	§2.4	**Role audit:** Independence and frequency of institutional audits, Cross-source information verification frequency, Active deviation behavior from algorithmic recommendations. **Affect audit:** Durability and irreplaceability of intimate relationships	Ordinal • Degradable
Human Biological Substrate (E_c)	§2.5	Conscious information throughput ($\sim$ 10 bits/s), Working memory capacity (4 ± 1 chunks)	Approximate Constant
Legitimacy Base ($\|L\|$)	§2.6	Rule of law index, Physical privacy protection coverage, Institutional trust surveys (regulatory response lag belongs to governance-cycle analysis, see Table 2)	Ordinal • Slow Accumulation, Rapid Collapse

When operationalizing meta-audit capacity into observable indicators, Nelson's (2009) model of professional skepticism can serve as a multi-factor structural reference: evidence, knowledge, traits, and incentives jointly shape audit judgments and actions. In the context of complex-estimate auditing, identifying contradictory evidence, integrating information across sources, and forming independent estimates may be listed as candidate observational directions. These indicators are scenario-specific operational choices made in this book, not a scale supplied by Nelson to validate MCAF's product structure. Nelson also notes that excessive professional skepticism can produce inefficiency and high costs. Proxy indicators should therefore measure whether calibration is appropriate rather than treating a unilateral increase in skepticism as an increase in meta-audit capacity.

The above proxy indicators provide empirical anchors for relative ordering, direction of change, and structural relations at the meta-framework level; they are not universal cardinal measures across scenarios. Without scenario-specific scales and aggregation rules, the six variables cannot be substituted directly into the meta-formula to calculate an absolute value of cognitive debt. Once those definitions are supplied, a scenario model may estimate local relations quantitatively. The proxy indicators provide directional criteria and falsifiable empirical anchors for projection mapping.

§7.5 A Minimally Theory-Laden Observation Interface

Even if the reader does not accept the term "cognitive debt," the following set of observable indicators, independent of the framework's terminology, provides a minimum consensus interface for empirical judgment. These indicators have a lower theoretical load than the meta-formula, but can independently trigger alertness:

1. **Is the verification window continuously shortening?** Is the time window between a user's contact with a product and their making an irreversible decision expanding or narrowing?
2. **Is the applicable boundary being continuously extrapolated?** Is the distance between the initially promised functional scope of the tool and its currently claimed functional scope widening?

3. **Who absorbs the failure cost?** When the tool malfunctions, are the consequences primarily borne by the projecting party (recall, compensation), or by the user or the public (health loss, attention waste, trust erosion)?
4. **Are benefit and responsibility decoupled?** Is the projecting party harvesting benefits while transferring risk to the user through disclaimers or narrative packaging?
5. **Has external calibration been replaced by endogenous evaluation?** Has the determination of product quality shifted from an independent third party to the platform's own rating system?
6. **Does a compensation loop exist?** When the tool produces negative consequences, does the user have institutional channels for remedy—return, appeal, independent arbitration—or is the only option "just don't use it"?
7. **Is the tool augmenting the user's judgment, or replacing it?** Is the tool accelerating the user's actions, or is it forming the user's intentions? When using a calculator, "to calculate 13×47" is decided by the user themselves—the calculator only accelerates the computational process; the purpose is formed by the user, and the tool only serves execution. In recommendation feeds optimized for engagement metrics such as dwell time, part of the judgment through which users form the intention "what do I want to see" is delegated to the algorithm. The algorithm generates a content stream from behavioral data, with dwell-time maximization as its optimization objective—the judgment of "what I want to see" is thereby outsourced as well. This shifts intention formation from something initiated entirely by the user to something co-shaped by the user and the platform. The relevant question is not whether users still form intentions, but the extent to which they continue to form, examine, and revise those intentions independently.

This set of indicators does not depend on any special terminology of the MCAF framework. Even if "cognitive debt" were replaced by "insufficiently tracked systemic consequences," the above seven points would still hold. They provide a minimum plane of consensus for readers who do not accept this framework: if multiple indicators among the above simultaneously worsen in a certain technological system, then regardless of whether the term "cognitive debt" is used, there is reason to believe that the system is generating some structural cost that is not being sufficiently booked.

§7.6 Boundary Statement on Quantitative Development

The MCAF framework allows quantitative development, but such development must proceed in a stratified manner and cannot skip levels. Below, three operational layers are numbered with A/B/C to distinguish them from the three-level epistemological demarcation of §7.2.

Layer A: Bookkeeping Discipline Layer. The meta-formula itself is a normative compulsory questioning structure and does not require quantitative closure. It only compels the analyst not to record only the projection-side gains without recording the defense-side costs. This layer is non-falsifiable (§7.2 Level 1) and does not need quantification.

Layer B: In-Scenario Indicator Layer. After binding the variables of the meta-formula to proxy indicators within a specific scenario, several sub-modules already possess quantitative potential. The media conductivity coefficient (Λ) naturally approximates measurable objects such as information arrival density, distribution speed, and recommendation penetration rate; unlocked stock (C) can use computing power capital expenditure (an input proxy, to be distinguished from stock), energy dispatch, and training data scale as approximate proxies; the narrative leverage ratio (Ω) is suitable for development into a risk ratio tool; the time-scales of the three clocks

are naturally suited for event time, response delay, and crisis trigger analysis. This layer is the most suitable for quantification and can be developed into indices, risk scores, threshold monitoring, and early warning tools.

Layer C: Cross-Scenario Meta-Framework Layer. As a framework spanning scenarios, the meta-formula permits structural comparison only after each scenario has defined and standardized its scales—for example, comparing relative levels of narrative pressure or types of clock ordering. Unstandardized data from different scenarios cannot be substituted directly into a single closed equation. At the meta-framework level, or across scenarios without standardization, directly substituting the six variables into the meta-formula to obtain an absolute numerical value of cognitive debt is explicitly prohibited.

If these three layers are not kept separate, a typical misreading will appear: what was originally a structural ledger is taken as a unified physical-type master equation. The true direction of development for this framework is not Newtonian-mechanics-style quantification, but is closer to financial risk measurement, stress testing, early warning indicators, state-space models, and latent variable estimation—a stratified, scenario-constrained quantitative toolkit centered on risk assessment and structural comparison. This is stronger than "purely heuristic," and more honest than "already a quantitative theory."

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§8 Conclusion: Discretization as Meta-Operation

The same structural act running throughout the entire chapter is discretization—discretization itself is not the lesion, but the usurpation of the outcomes of discretized instrumental rationality as the substance of value rationality is the root cause of the erosion of the defense line.

The previous chapter has already laid the ground for this judgment from the epistemological level: mapping is systematically lossy, and the act of mapping transfers the cognizer from the time dimension to the clock dimension. This chapter has shown the operational unfolding of the same judgment from the level of the bookkeeping structure: the meta-operation by which the projection side penetrates the defense side is discretization.

Von Neumann established a cognitive-physiological basis for this rupture: the neural message system has a statistical character, while conscious mathematics and logic are secondary languages built upon it; the primary language and explicit symbolic forms are not simply isomorphic (von Neumann, 1958). Zhao Dingxin explained the lossy compression at the institutional level—the loss of the depth of reality due to the extraction of discretized variables by the hermeneutic circle (Zhao Dingxin, 2022). Goodhart's law and Campbell's law revealed the endogenous corrosion caused after continuous principles of justice are discretized into a checklist of tick-box compliance items—procedural legitimacy degrades from topological judgment to metric calculation; the anchoring paradox is precisely the systemic pathology manufactured by the operation of discretization within the legitimacy base.

The role/affect dichotomy of meta-audit capacity marks the ontological boundary of discretization. Role audit is algorithmically comprehensible—compliance, efficiency, procedural justice can all be formalized and optimized. Affect audit is incommensurable between human and machine—anchored in intimate bonds, it cannot be encoded, cannot be outsourced. The structural effect of technological optimization is precisely this: by infinitely optimizing role audit, it induces the subject to believe that the defense is still solid, thereby masking the decay occurring in the core zone of affect audit. Discretizing the person out from their relationships as an

analytical tool is effective, but treating the discretized atomistic individual as the substance of the person—this is the ultimate engineering form of what §5.3 calls algorithmic *xiangyuan*.

From this follows the core synthesis of this chapter: discretization itself is not the lesion; usurping the outcomes of discretized instrumental rationality as the substance of value rationality is the root cause of the collapse of the defense line.

What must be introspected is that the meta-formula of this chapter is itself an act of discretization—six variables, multiplication-subtraction operations, ordinal measures. §7.3 has already completed the self-audit on this: the MCAF framework is a clock, not the sun. The tool must be discrete, but insight must point to the continuous.[4]

The Roadmap of the Book. This chapter has established the core conceptual system of the MCAF framework. The subsequent chapters unfold using this framework: Chapter 4 formalizes the fiber framework, providing mathematical tools for the three layers of irreversibility; Chapter 5, grounded in Francis L. K. Hsu's cross-civilizational comparative anthropology, unfolds the eight layers of PSH as the ontological foundation of meta-audit capacity, demonstrates the differentiated vulnerability lines of two equilibrium ideals in the face of the clock's intrusion, and argues for the modulation function of the state as a historical extension of PSH; Chapter 6 analyzes the power structure of the monopolization of the mapping function; Chapters 7 and 8 operationalize the two derived indices of the cognitive Reynolds number and the narrative leverage ratio. Chapters 9 through 16 apply the meta-formula to AI scenarios; Chapters 17 through 22 apply it to financial scenarios, testing the falsifiability of the framework through projection mapping; Chapters 23 through 24 trace the history of knowledge production in academia and capital markets; Chapters 25 through 29 unfold the definition and genealogy of the Discrete Leviathan, and the economic foundations of control over distribution; Chapters 30 through 33 address commitment vehicles, agentic AI, the Great Acceleration, and the city; Chapters 34 through 38 converge on the history of route filtering, endgame conditions, the stalling of critique, and governance and action.

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[4]This judgment is a **structural diagnosis** of a feature of modernity, not a universal negation of the value of modernity. Its empirical validity will be tested in the scenario mappings of the subsequent chapters (AI, Finance, City and Real Estate).

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